PLSC 40601

Week 6: Policy learning and evaluation.

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Spring 2026

Why we experiment

The Washington Post
Democracy Dies in Darkness

ELECTIONS Live updates Election 2024 Nov. 7 results Democracy In America Midterms 2022

Biden team looks at buying more ads amid ongoing polling concerns

The president has told others that he remains concerned about his stubbornly low approval numbers in battleground states

By Michael Scherer and Tyler Pager
October 22, 2023 at 6:00 a.m. EDT

Why we experiment

The Biden campaign has drawn up options for expanding its unconventional $25 million early advertising effort this year, as the president continues to express frustration in private conversations about the state of his polling in battleground states, according to people familiar with the discussions.

Which ad(s) should the campaign run?

How do we design a study to answer this question?

To achieve this goal, the experimenter faces three tasks.

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1. **Treatment allocation procedures.** How to collect data during the experiment.

2. **Recommendation procedures.**

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1. **Treatment allocation procedures.** How to collect data during the experiment.

2. **Recommendation procedures.** Using the experimental data, how to select the "best" treatment(s)$\rightarrow$ optimal policy learning.

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1. **Treatment allocation procedures.** How to collect data during the experiment.

2. **Recommendation procedures.** Using the experimental data, how to select the "best" treatment(s)$\rightarrow$ optimal policy learning.

2. **Evaluation procedures.**

How do we design a study to answer this question?

To achieve this goal, the experimenter faces three tasks.

1. **Treatment allocation procedures.** How to collect data during the experiment.

2. **Recommendation procedures.** Using the experimental data, how to select the "best" treatment(s)$\rightarrow$ optimal policy learning.

2. **Evaluation procedures.** Using the experimental data, how to evaluate the learned policy.

How do we design a study to answer this question?

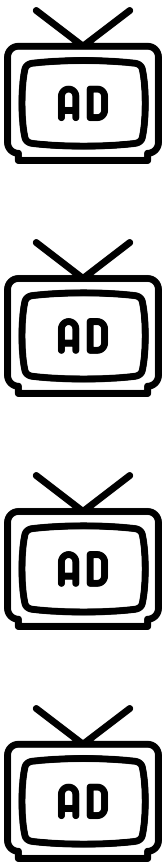
Advertisements		(Unknown) mean response
	A	θ_A
	B	θ_B
	C	θ_C
	D	θ_D

How do we design a study to answer this question?

Advertisements	(Unknown) mean response
A	0.44
B	0.44
C	0.45
D	0.50

Uniform allocation

Advertisements



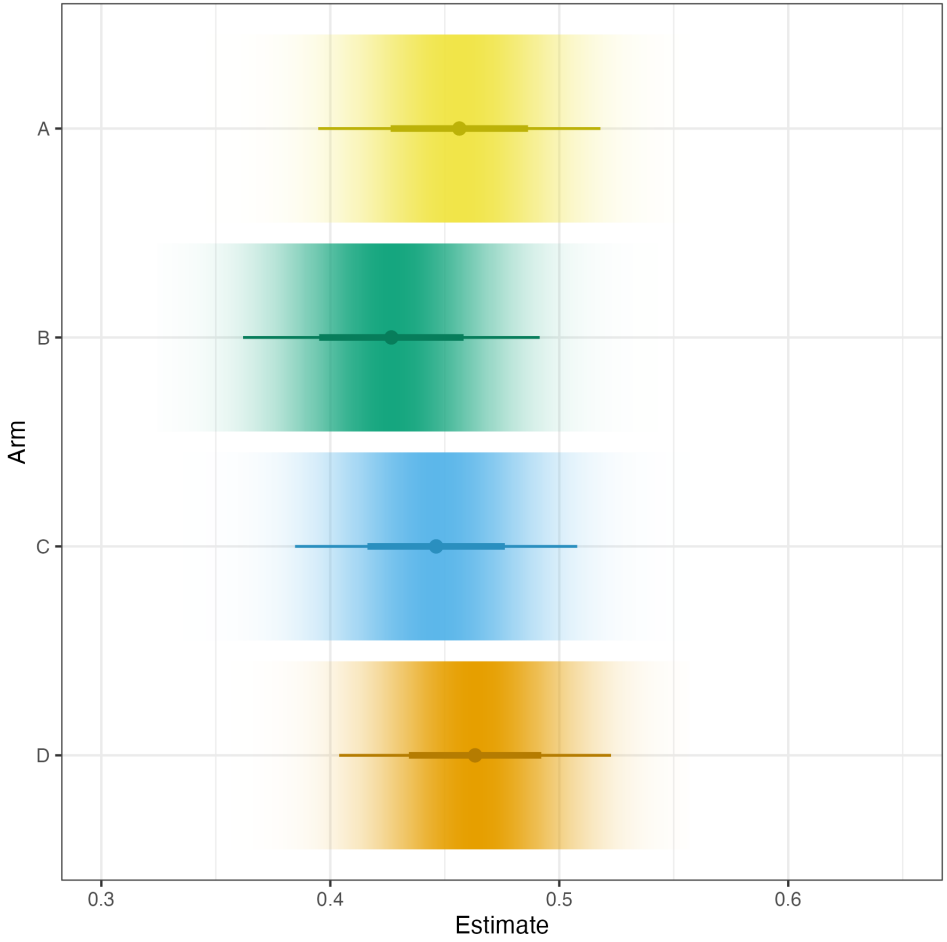
A

B

C

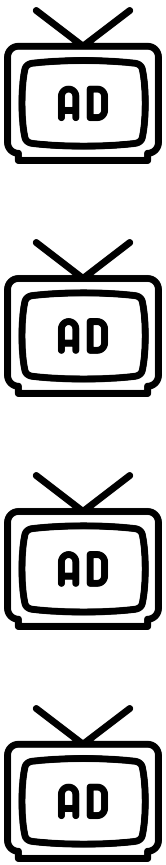
D

Estimated
mean response



Uniform allocation; EBA recommendation

Advertisements



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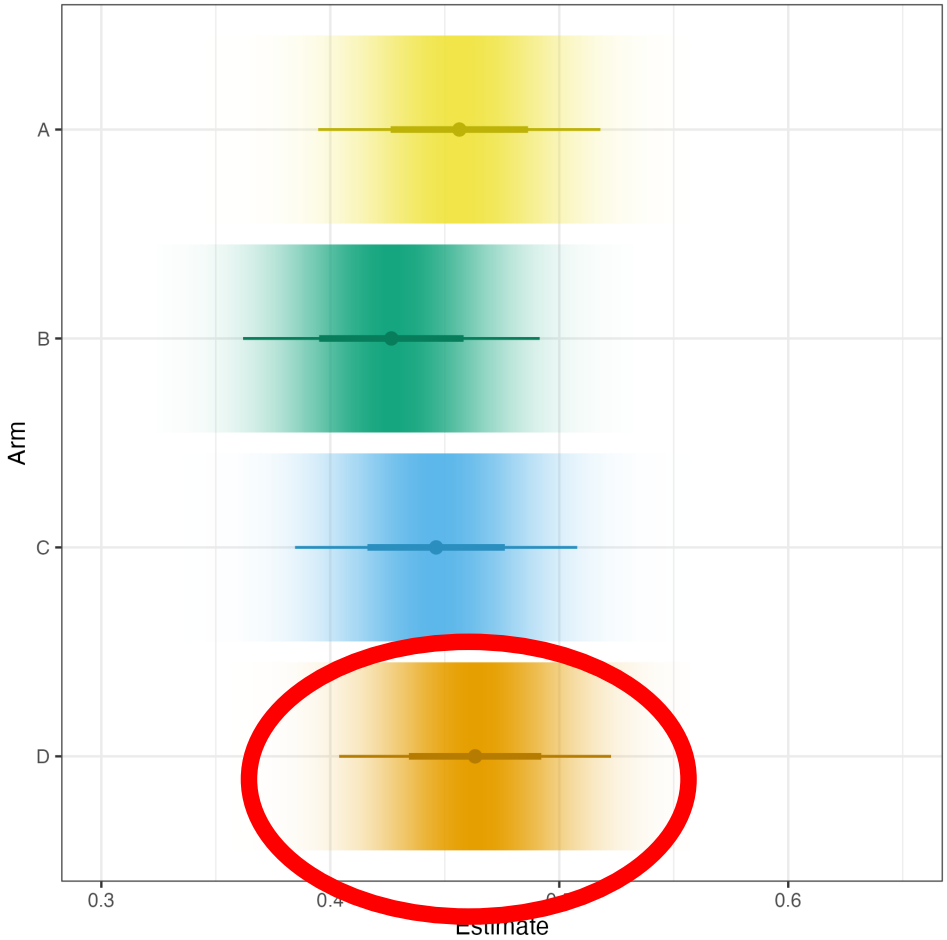

Did we answer our research question?

Framework

- Individuals indexed $i = 1, \ldots, n$

Framework

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- Treatments $W_i \in \mathcal{W}$, where $|\mathcal{W}| = k$

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- Individuals indexed $i = 1, \ldots, n$
- Treatments $W_i \in \mathcal{W}$, where $|\mathcal{W}| = k$
- Potential outcome $Y_i = Y_i(W_i) \in \mathbb{R}$;
  vector of potential outcomes $\mathbf{Y}_i = (Y_i(1), \ldots, Y_i(k))$
- i.i.d. $\mathbf{Y}_i$

Evaluation metric

Best arm:

$$w^* := \arg\max_w \mathrm{E}\left[Y_i(w)\right].$$

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In practice, we select $\hat{w}$ based on our allocation and recommendation procedures.

Evaluation metric

Best arm:

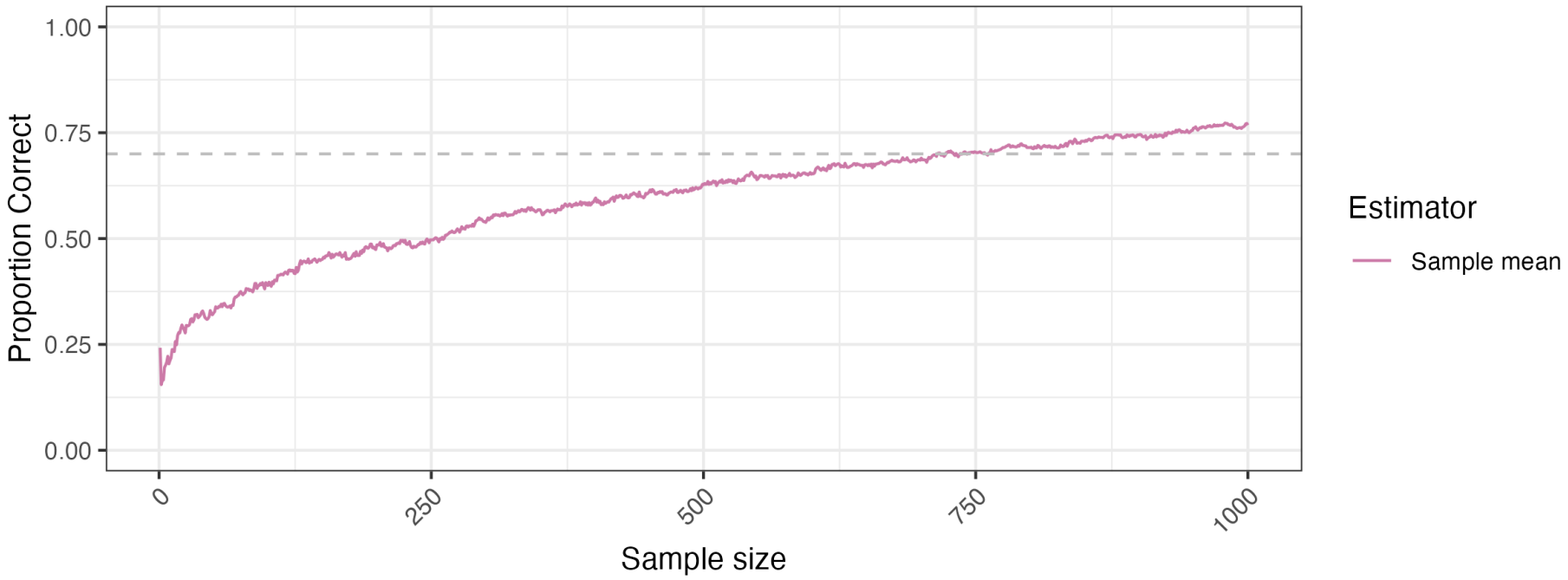
$$w^* := \arg\max_{w} \mathrm{E}\left[Y_i(w)\right].$$

In practice, we select $\hat{w}$ based on our allocation and recommendation procedures.

Evaluate:

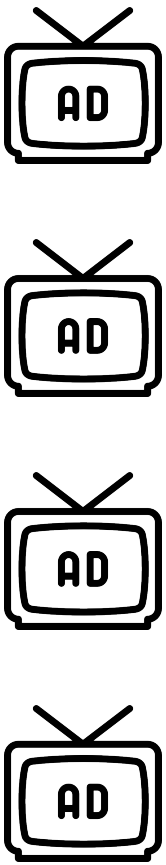
$$\mathrm{E}\left[\hat{w} = w^*\right].$$

Evaluation metric



Can we better allocate our data?

Advertisements



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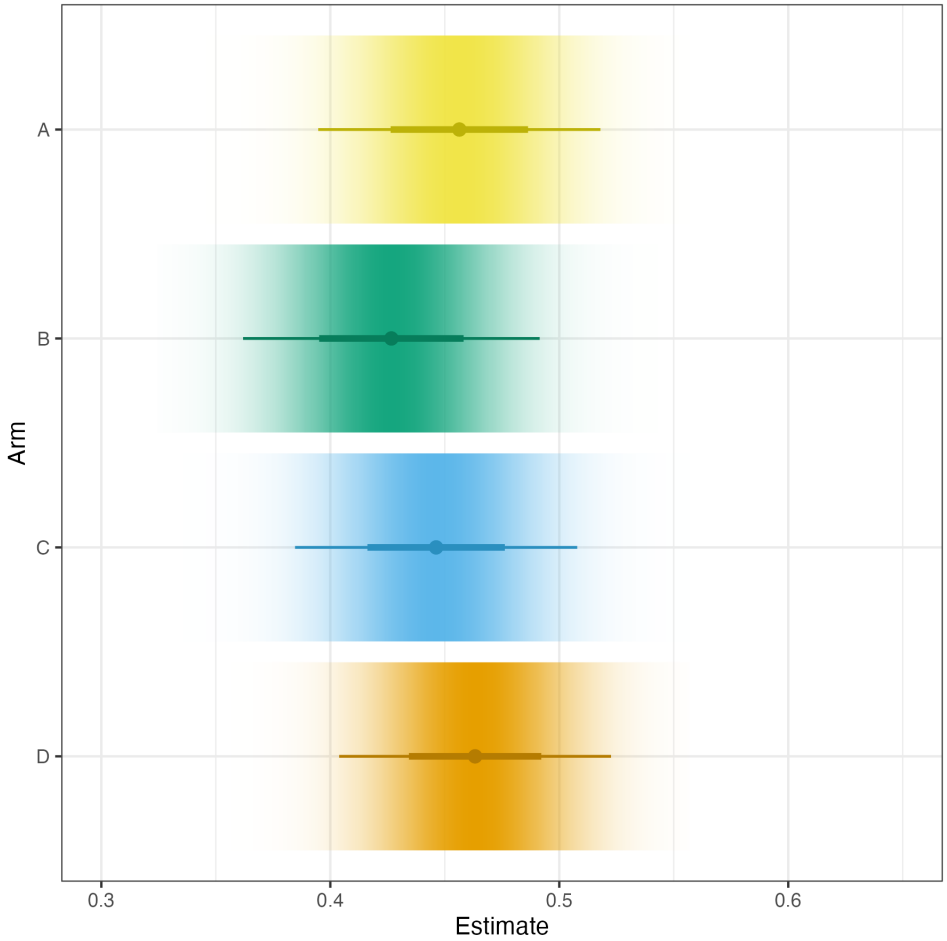
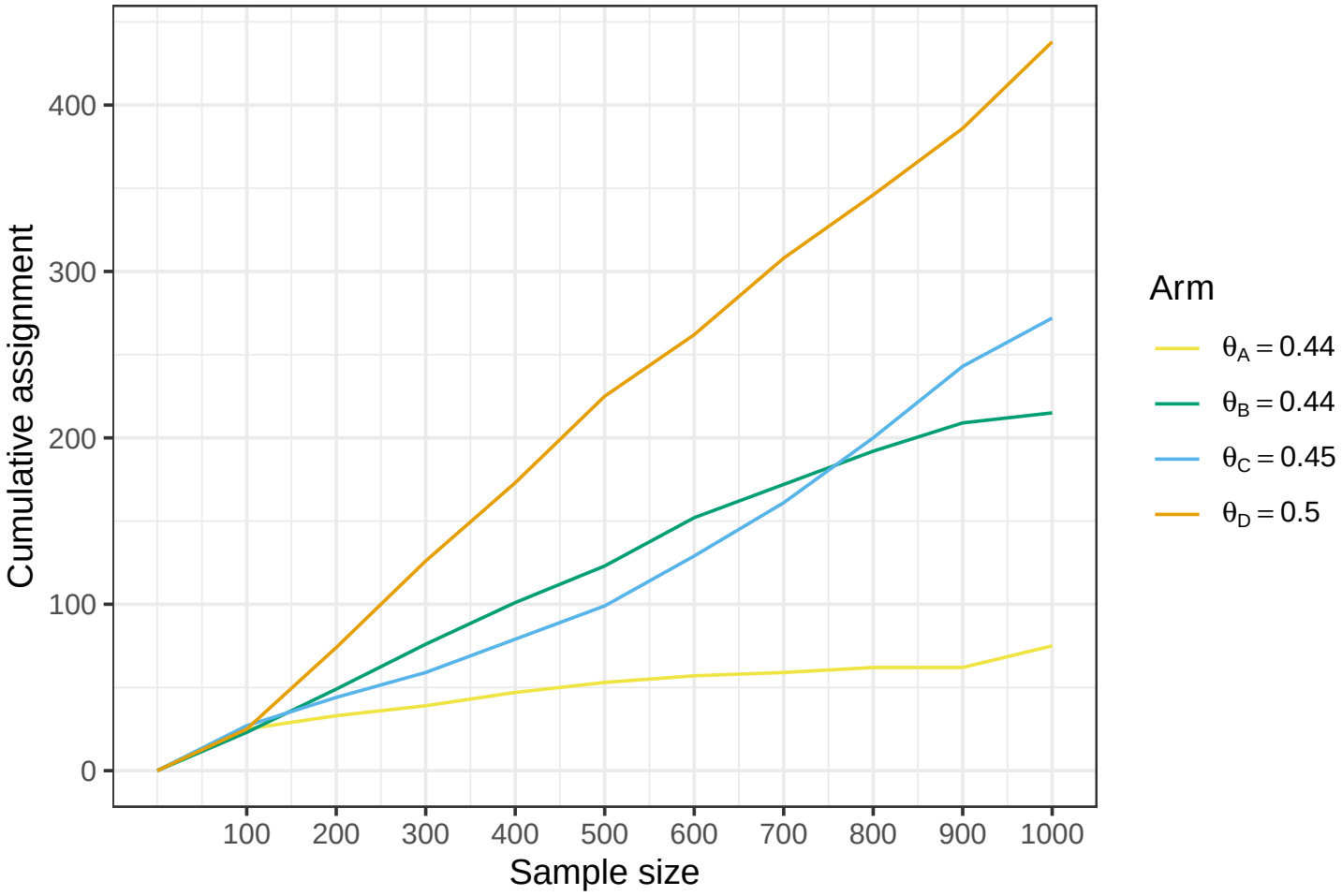


Response adaptive allocation



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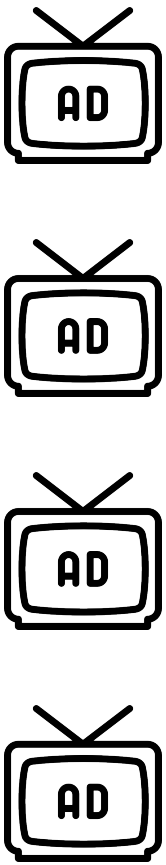

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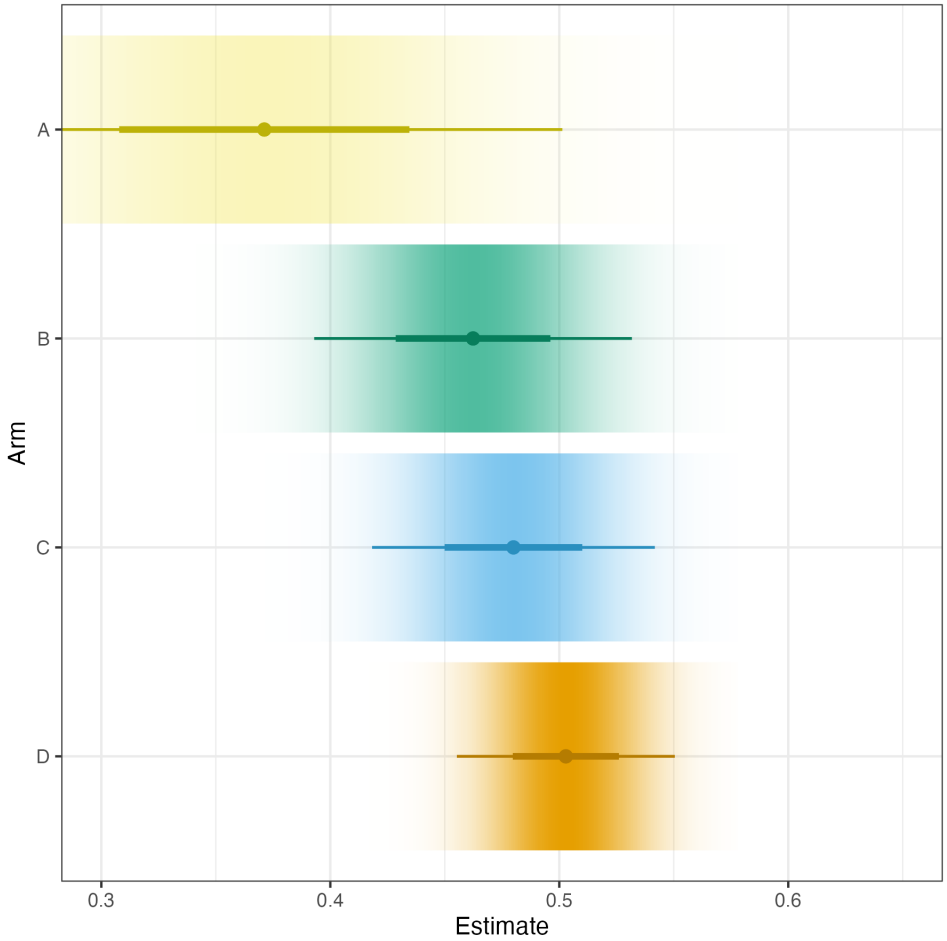

Framework

- Account for some **history** at time we see individual i:

$$\mathbf{H}_i = \begin{bmatrix} Y_1 & W_1 & X_{[1]1} & \cdots & X_{[1]p} \\ Y_2 & W_2 & X_{[2]1} & \cdots & X_{[2]p} \\ \vdots & & & & \vdots \\ Y_{i-1} & W_{i-1} & X_{[i-1]1} & \cdots & X_{[i-1]p} \\ \vdots & & & & \vdots \\ ? & ? & ? & \cdots & ? \end{bmatrix}$$

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- We don't assume the whole sampling frame is known.

Framework

- Treatment assignment probability

$$e_i(w) := \Pr\left[W_i = w | \mathbf{X}_i = x\right]$$
$$\mathbf{e} := (e(1), \ldots, e(k))$$

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- **Policy**

$$\pi : \mathbf{H}, \mathbf{X} \to \mathbf{e},$$

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- **Policy**

$$\pi : \mathbf{H}, \mathbf{X} \to \mathbf{e},$$

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    (functional form, cost constraints in assignment probabilities)

Thompson sampling

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Setup: Multiple treatment arms. Start with uniform priors.

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Timepoint 1:

- Assign treatment uniformly at random.
- Observe response.
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Timepoint 2:

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$\vdots$

Repeat actions from batch 2.

$\vdots$

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⋮

Repeat actions from batch 2.

⋮

Stopping rule

Thompson sampling

- Assign treatment according to posterior probability that each arm is best.

Thompson sampling

```
> set.seed(40601)
> # true parameter values
> thetas <- c(0.45, 0.47, 0.48, 0.5)
> # simple random assignment
> (n <- table(factor(sample(1:4, 10, replace = TRUE), levels = 1:4)))

1 2 3 4
2 4 3 1

> # successes, based on thetas
> (successes <- mapply(rbinom, 1, size = n, prob = thetas))

[1] 1 2 2 1

> # respective posterior distributions, based on # successes, failures
> (shapes <- rbind(successes = successes, failures = n-successes))

          1 2 3 4
successes 1 2 2 1
failures  1 2 1 0

> # sample from posteriors
> outmat <- mapply(rbeta, 1000,
+                  shape1 = 1 + shapes[1,],
+                  shape2 = 1 + shapes[2,])
> # proportion of times posterior samples in each group were highest
> (probs <- prop.table(table(apply(outmat, 1, which.max))))

    1     2     3     4
0.147 0.118 0.281 0.454
```

Policies: response adaptive allocation

$$\pi : \mathbf{H}, X \rightarrow \mathbf{e}$$

Examples of policies: Response-adaptive allocation

					$\mathbf{H}_i$		
Unit	$e_i(A)$	$e_i(B)$	$e_i(C)$	$e_i(D)$	Y_i	W_i	$X_{[i]1}$
1	0.25	0.25	0.25	0.25	?	?	?
2	?	?	?	?	?	?	?
3	?	?	?	?	?	?	?
4	?	?	?	?	?	?	?
5	?	?	?	?	?	?	?
6	?	?	?	?	?	?	?

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2	?	?	?	?	?	?	?
3	?	?	?	?	?	?	?
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2	?	?	?	?	?	?	?
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3	?	?	?	?	?	?	?
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3	0.27	0.26	0.27	0.20	?	?	?
4	?	?	?	?	?	?	?
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3	0.27	0.26	0.27	0.20	?	A	?
4	?	?	?	?	?	?	?
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3	0.27	0.26	0.27	0.20	0	A	?
4	0.11	0.32	0.32	0.26	?	?	?
5	?	?	?	?	?	?	?
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4	0.11	0.32	0.32	0.26	1	C	?
5	0.07	0.25	0.50	0.18	?	?	?
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4	0.11	0.32	0.32	0.26	1	C	?
5	0.07	0.25	0.50	0.18	0	A	?
6	0.03	0.26	0.52	0.19	?	?	?

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5	0.07	0.25	0.50	0.18	0	A	?
6	0.03	0.26	0.52	0.19	?	A	?

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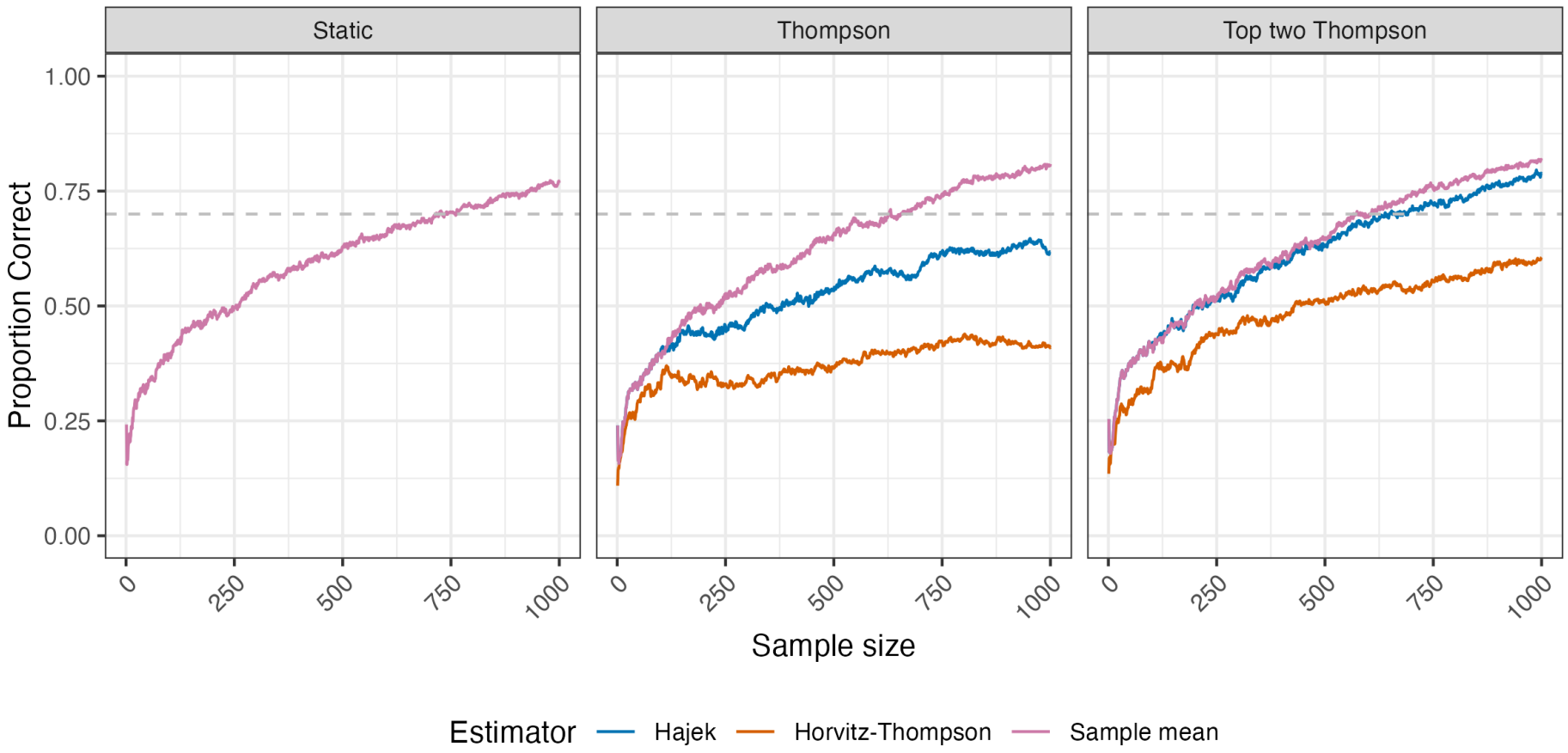
					$\mathbf{H}_i$		
Unit	$e_i(A)$	$e_i(B)$	$e_i(C)$	$e_i(D)$	Y_i	W_i	$X_{[i]1}$
1	0.25	0.25	0.25	0.25	0	D	?
2	0.30	0.30	0.30	0.10	1	D	?
3	0.27	0.26	0.27	0.20	0	A	?
4	0.11	0.32	0.32	0.26	1	C	?
5	0.07	0.25	0.50	0.18	0	A	?
6	0.03	0.26	0.52	0.19	0	A	?

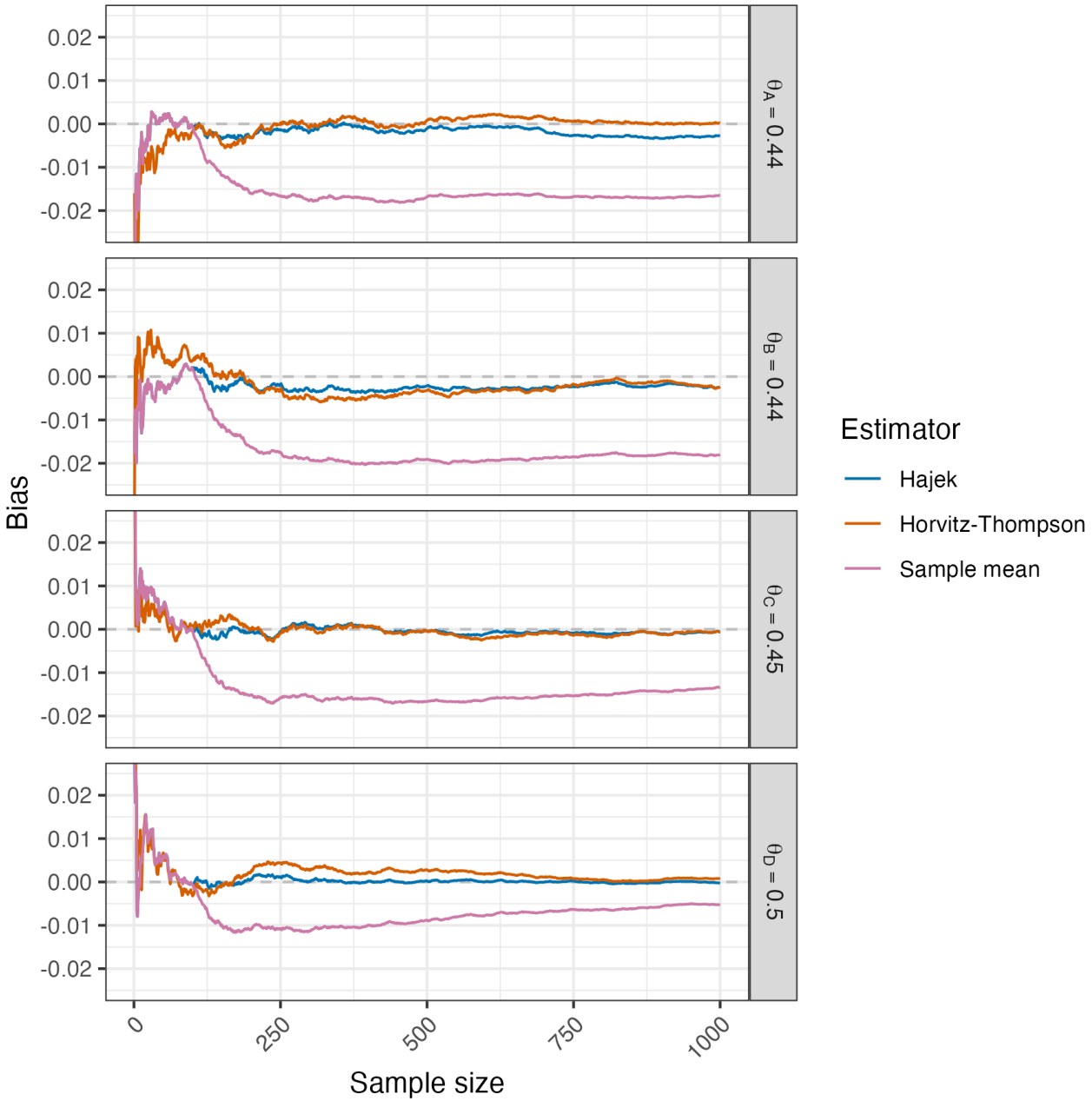
Top-two Thompson sampling (Russo, 2016)

- $\rightarrow$ Sample from posterior probabilities;
- $\rightarrow$ play arm with the highest value with probability B;
- $\rightarrow$ resample;
- $\rightarrow$ play next arm with highest value with probability $1 - B$.

Evaluation metric



Estimating procedures with TT Thompson



Bias

Best arm identification

- Why these types of designs work:

Best arm identification

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  - Strategic *exploration* at the top end of the distribution

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  - Promising in the fixed-budget setting (Jourdan et al., 2022) with good simulation-based results (Russo, 2016)
  - Uniform random assignment will do well asymptotically
  - But we can get closer to best policy faster in small, medium sized samples.

A few more questions

- Do we care about frequentist inference ex-post?

Inference on data-driven quantities

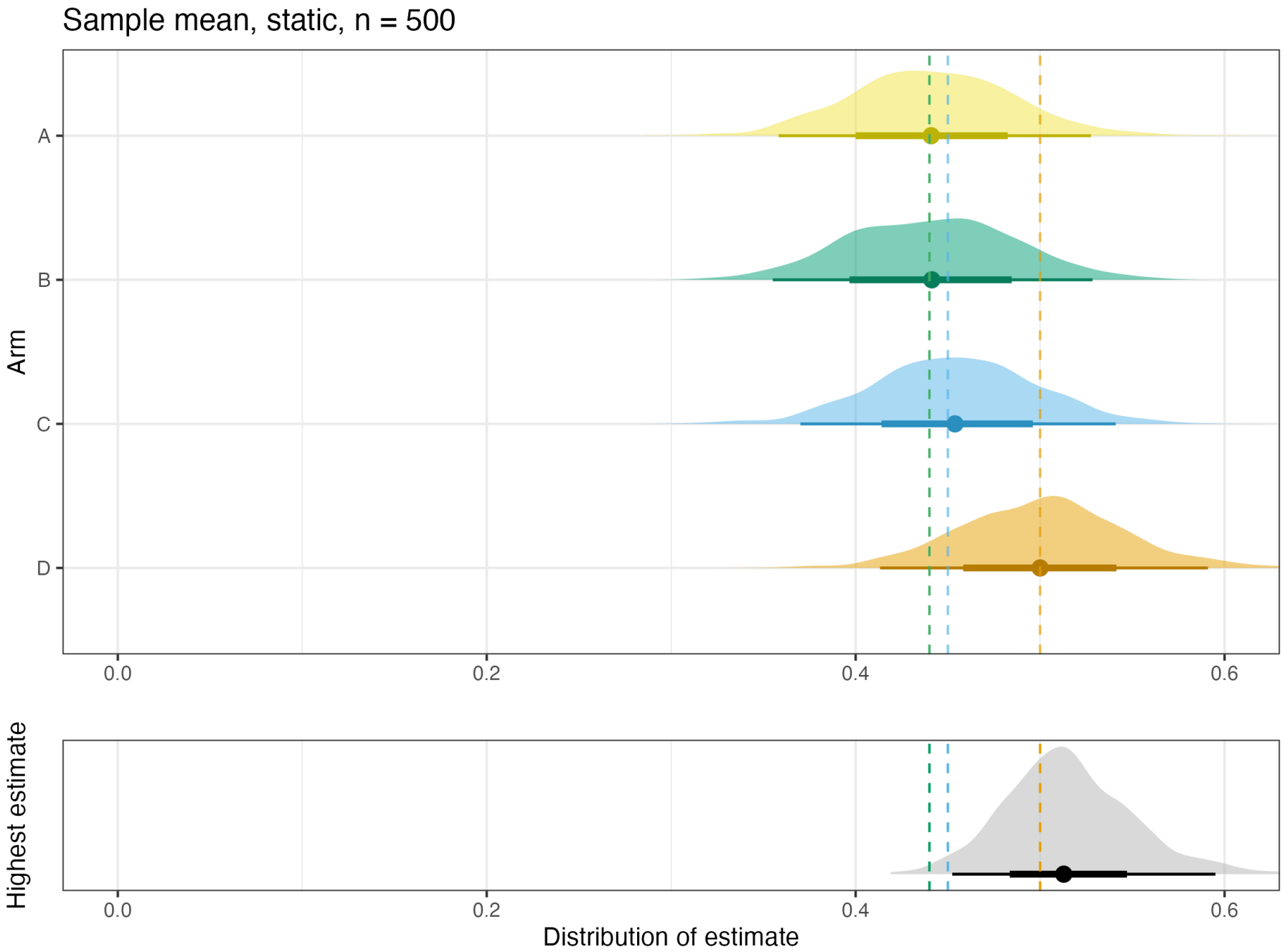
- Naive inference on “best” arm is positively biased

Inference on data-driven quantities

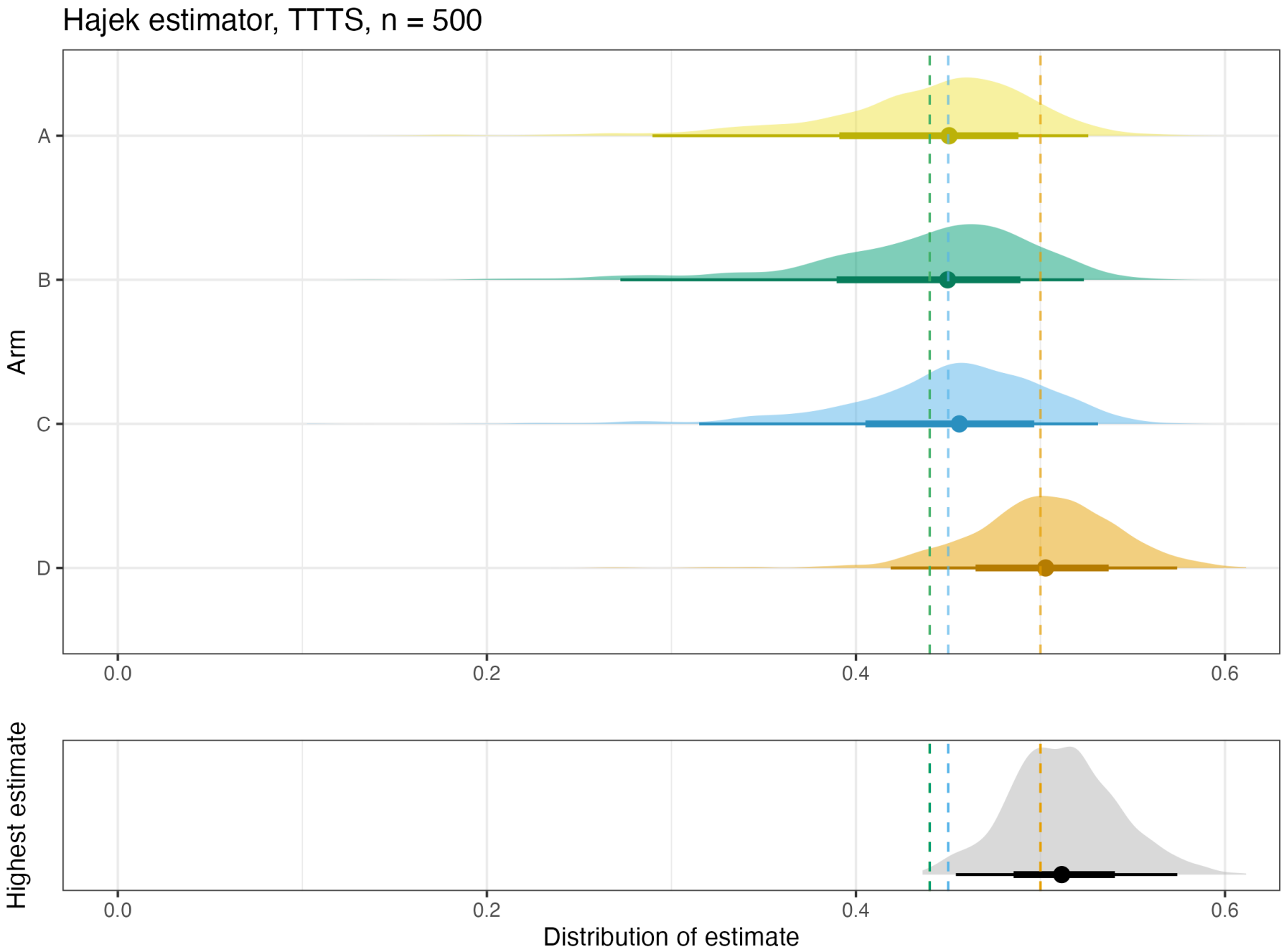
- Naive inference on “best” arm is positively biased

$\rightarrow$ version of winner’s curse

Inference on data-driven quantities



Inference on data-driven quantities



Inference on data-driven quantities

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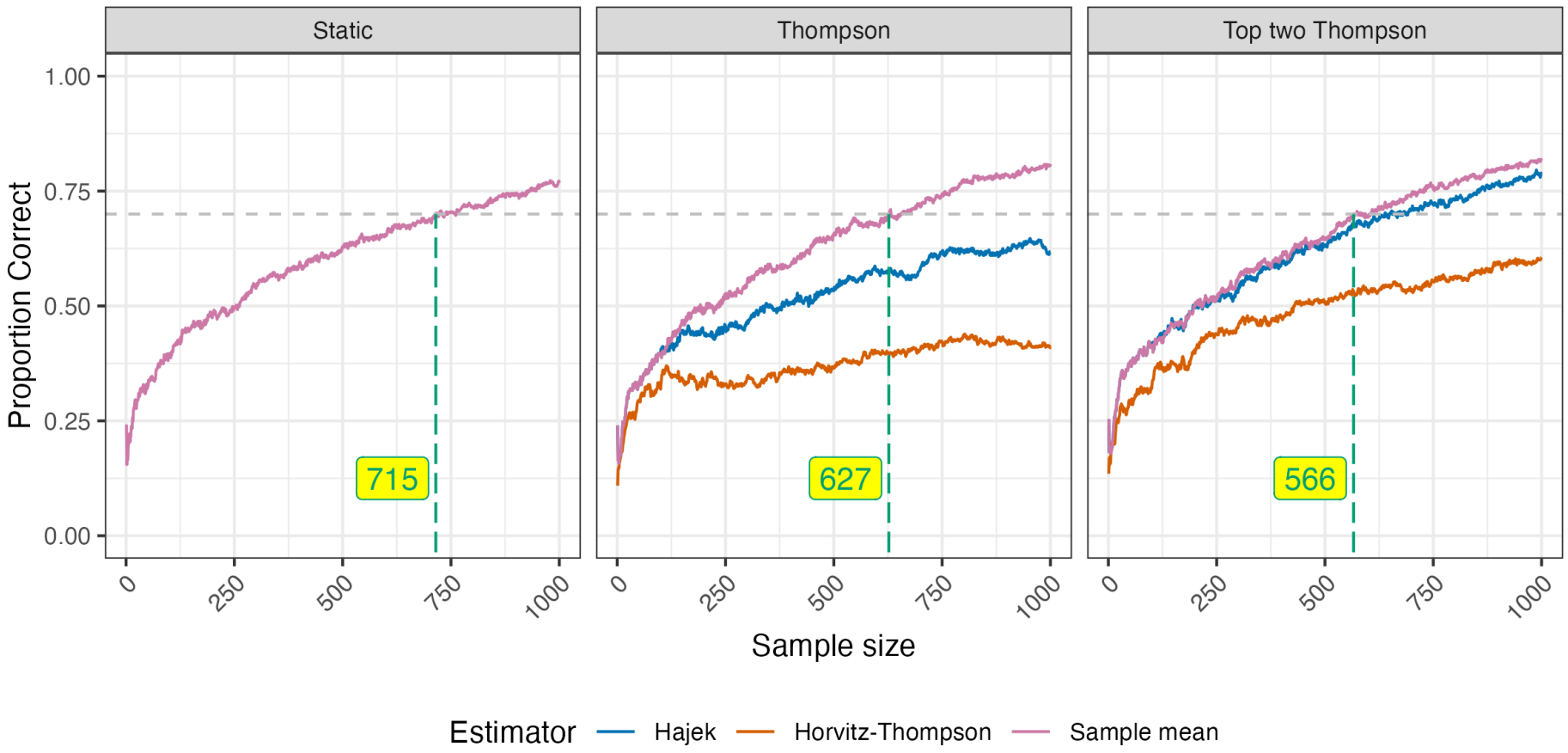
→ Some tools for selective inference on non-adaptively collected data (Andrews et al., 2024)

- Or use cross-fitting

- For adaptively collected data, without further assumptions on experiment design, just conduct inference using last batch of data collected (Chen and Andrews, 2023)

Evaluation metric



Confidence intervals with adaptively collected data

Even for non data-driven hypotheses, with adaptively collected data, we get non-normality in distribution of estimators.

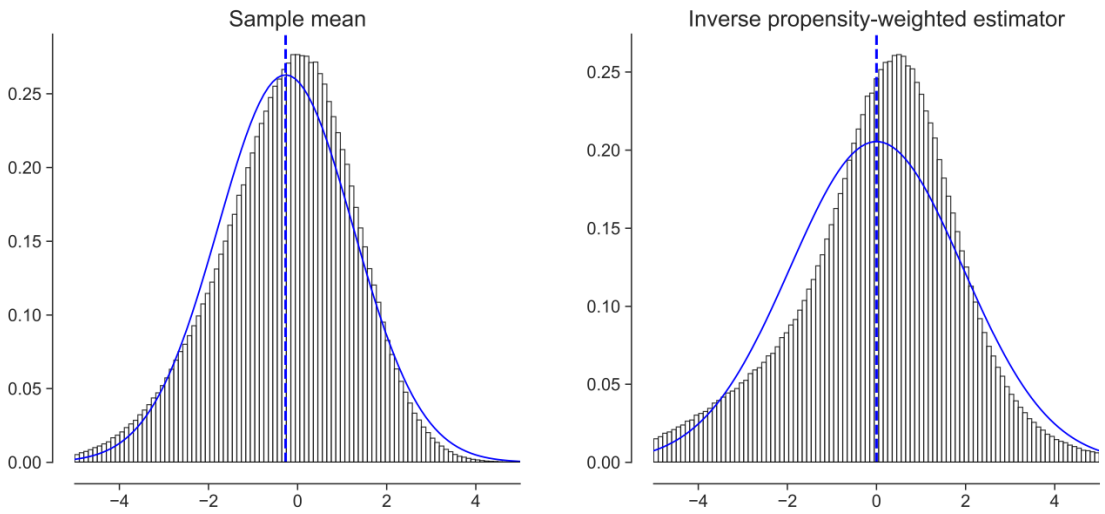
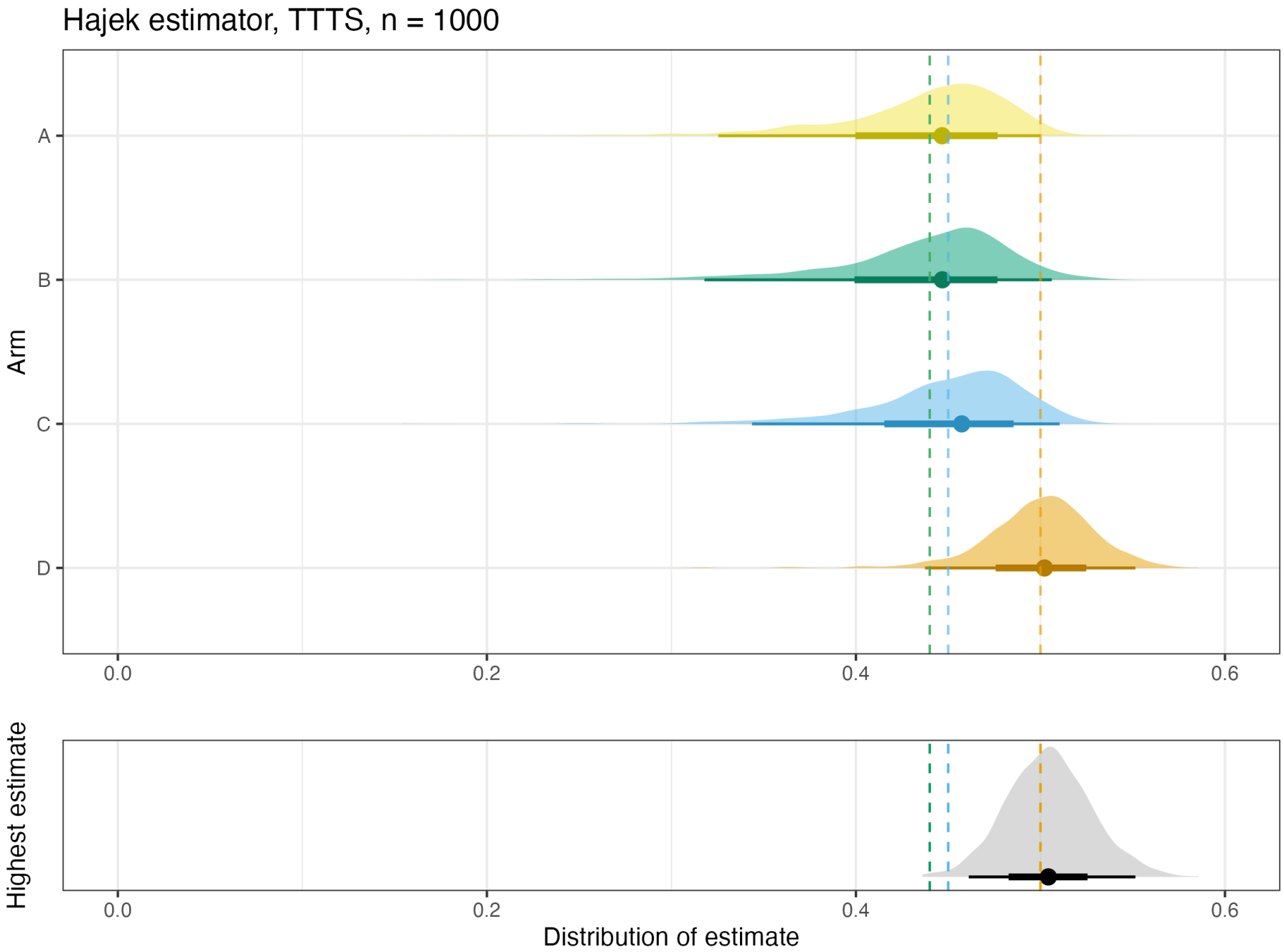


Figure 1: Distribution of the estimates $\widehat{Q}^{\mathrm{AVG}}(1)$ and $\widehat{Q}^{\mathrm{IPW}}(1)$ described in the introduction. The plots depict the distribution of the estimators for $T = 10^6$, scaled by a factor $\sqrt{T}$ for visualization. The distributions are overlaid with the normal curve that matches the first two moments of the distribution, along with a dashed line that denotes the mean. All numbers are aggregated over 1,000,000 replications.

Hadad et al. (2021)

Inference on data-driven quantities



Confidence intervals with adaptively collected data

Some solutions:

- Hadad et al. (2021): adaptively re-weight the terms of the augmented inverse propensity weighting estimator, controlling for contributions of each term to estimator's variance.
  - Requires that the propensity score decay at a slow enough rate.
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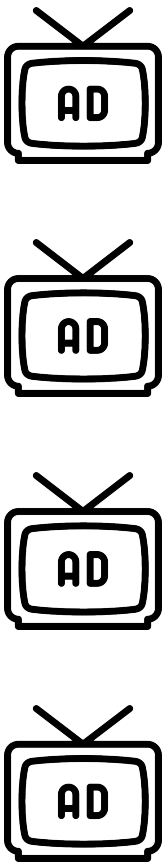
- Deshpande et al. (2017): augment OLS estimator with $\mathbf{W}-$ "decorrelation" matrix to remove the effects of adaptivity.
- Zhang et al. (2020): batched OLS hypothesis testing; OLS estimates are asymptotically normal *within* batches. Then combine batch-wise statistics and compare to simulated null distribution
  - Requires batched assignment.

Confidence intervals with adaptively collected data

- You still need an appropriate design (sufficient probability floors, batch-wise assignment) for these methods to work.

Confidence intervals with adaptively collected data

Advertisements



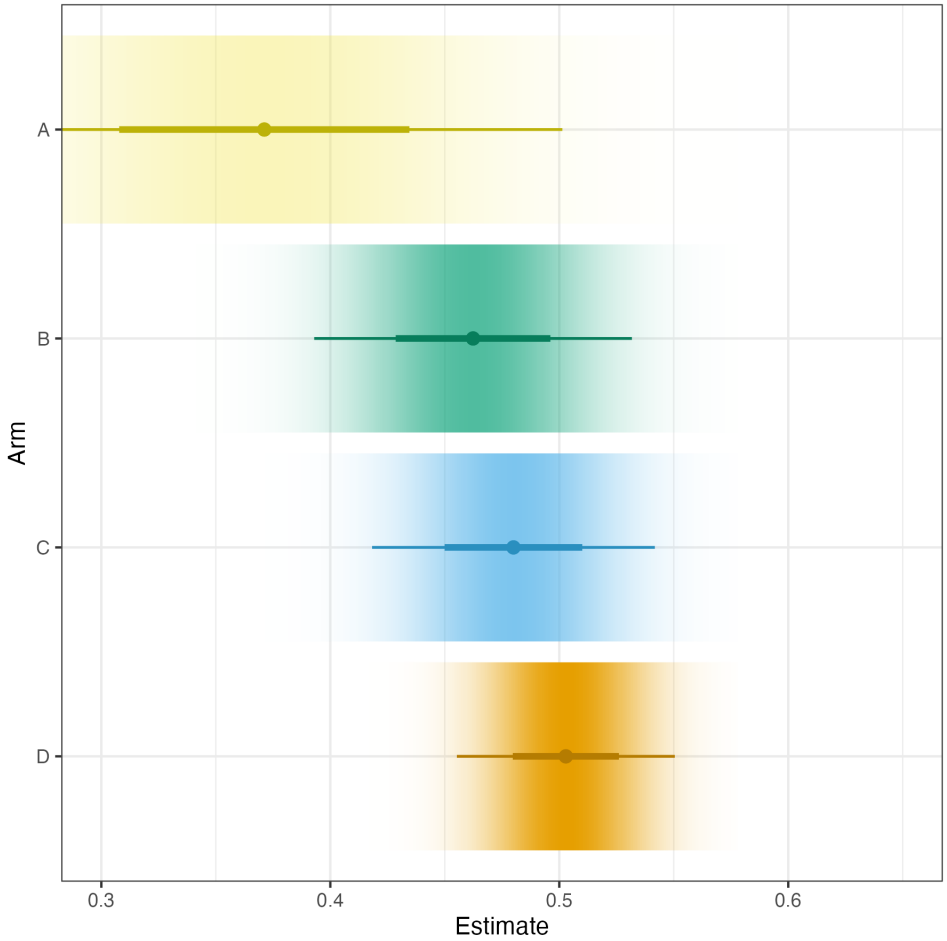
A

B

C

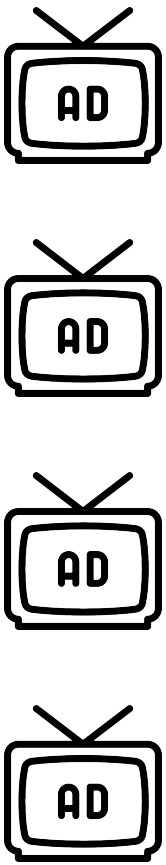
D

Estimated
mean response



Confidence intervals with adaptively collected data

Advertisements



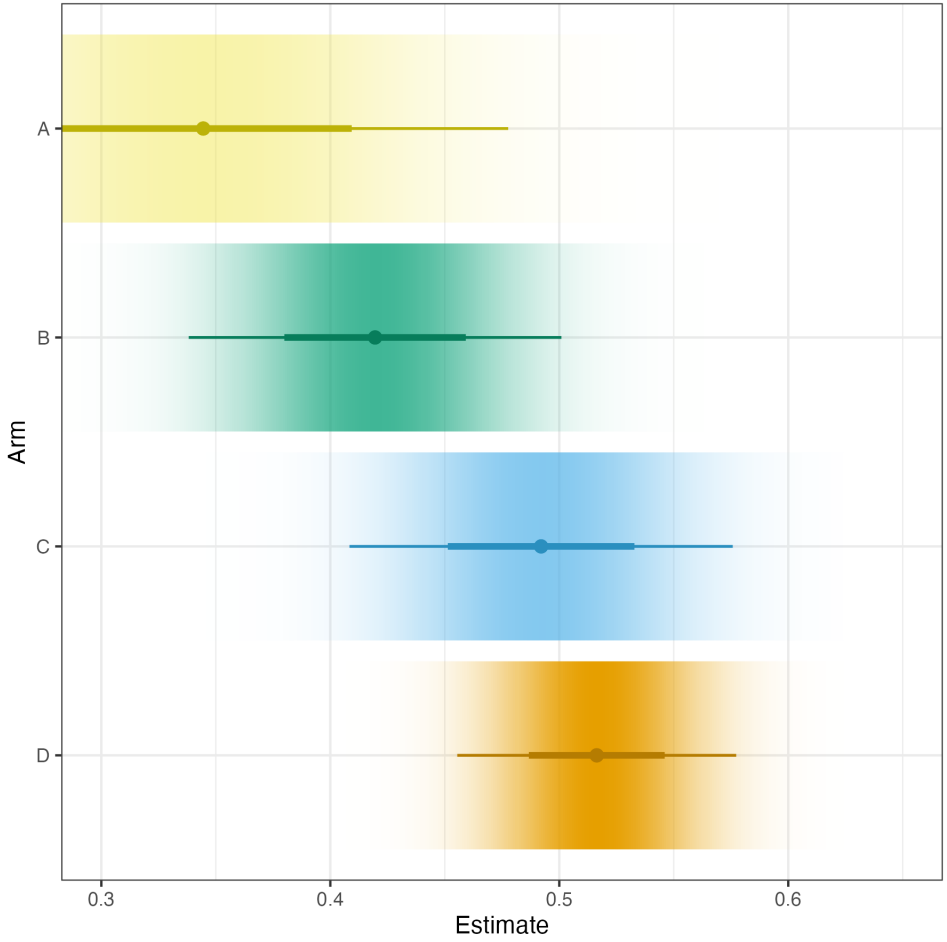
A

B

C

D

Adjusted estimated mean response



Confidence intervals with adaptively collected data

`banditsCI` package

Should the campaign buy ads at all?

Why we experiment

The Biden campaign has drawn up options for expanding its unconventional $25 million early advertising effort this year, as the president continues to express frustration in private conversations about the state of his polling in battleground states, according to people familiar with the discussions.

Biden signed off on the initial fall television and digital campaign despite concerns from some advisers who felt that the money could be better allocated for other priorities like building up staff sooner or growing cash reserves. Other Democrats continue to question whether the upbeat sales pitch in some Biden ads fails to reflect the economic realities of voters.

Why we experiment

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$$\pi^* = \arg\max_{\pi \in \Pi} \mathrm{E}\left[\langle \mathbf{Y}_i, \pi(\mathbf{X}_i) \rangle\right]$$

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$$\begin{aligned} \mathrm{H}_0 :& \mathrm{E}\left[\langle \mathbf{Y}_i, \pi(\mathbf{X}_i) \rangle\right] = 0.45 \;\; \forall \pi \in \Pi \\ \mathrm{H}_A :& \exists\, \pi \in \Pi : \mathrm{E}\left[\langle \mathbf{Y}_i, \pi(\mathbf{X}_i) \rangle\right] > 0.45 \end{aligned}$$

Why we experiment

For uniform policies:

$$\mathrm{H}_0 : \theta_w = 0.45 \;\; \forall w \in \mathcal{W}$$
$$\mathrm{H}_A : \exists \; w \in \mathcal{W} : \theta_w > 0.45$$

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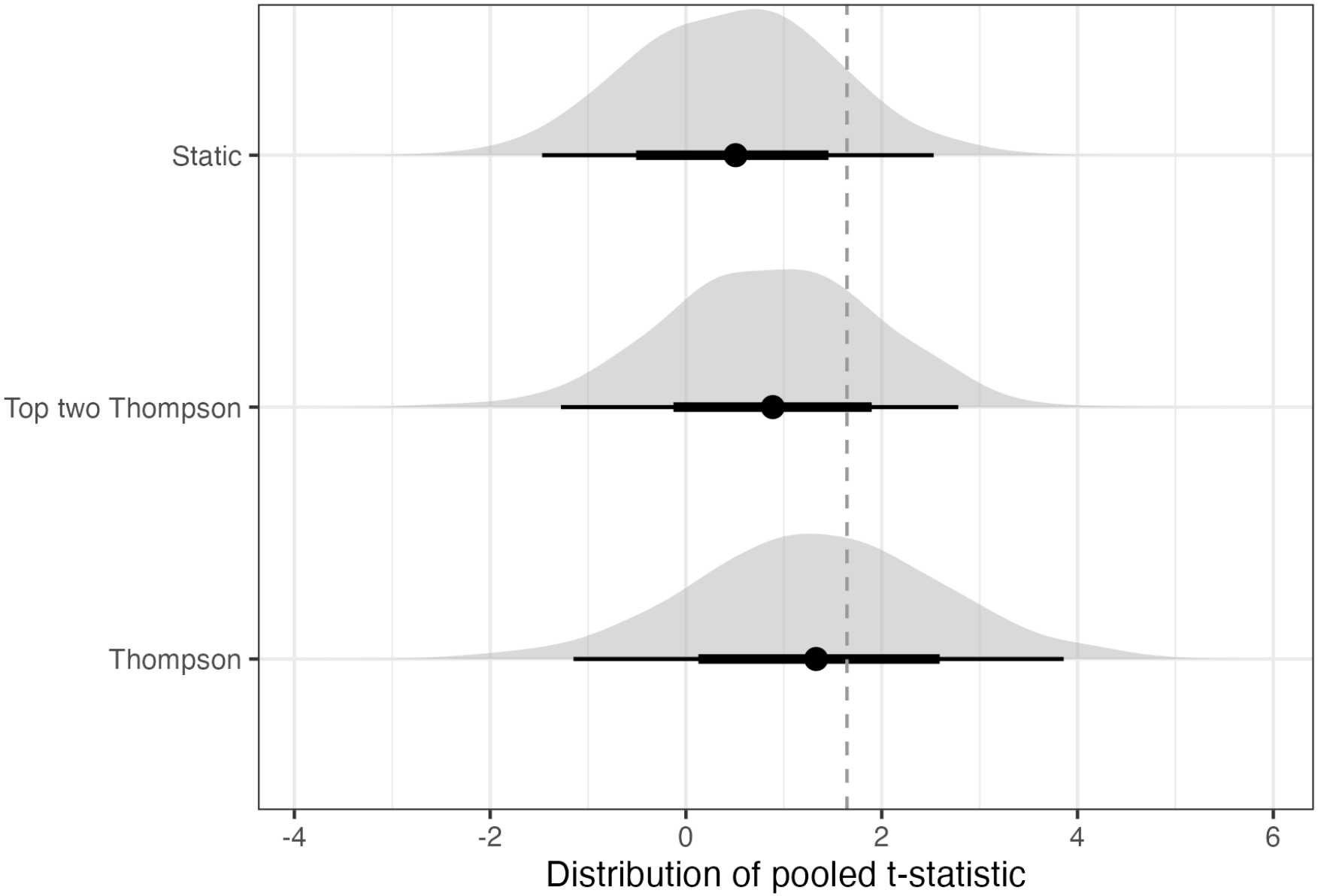
- It doesn't make sense to test this with pooled data in the static setting if some of the advertisements are worse than status quo
  - $\rightarrow$ Instead: test each message, implement adjustments for multiple testing procedures.

- For algorithms that minimize cumulative regret, assignment will move towards optimal allocation, and we can test directly whether experimental allocation is better than status quo.

Cumulative regret

$$R_{cumulative}(\tilde{\pi}) = \sum_{i=1}^{n} \left(\mathrm{E}\left[\langle \mathbf{Y}_i, \pi^* \rangle\right] - \mathrm{E}\left[\langle \mathbf{Y}_i, \tilde{\pi}(\mathbf{H}_i) \rangle\right] \right)$$

Inference on data-driven quantities



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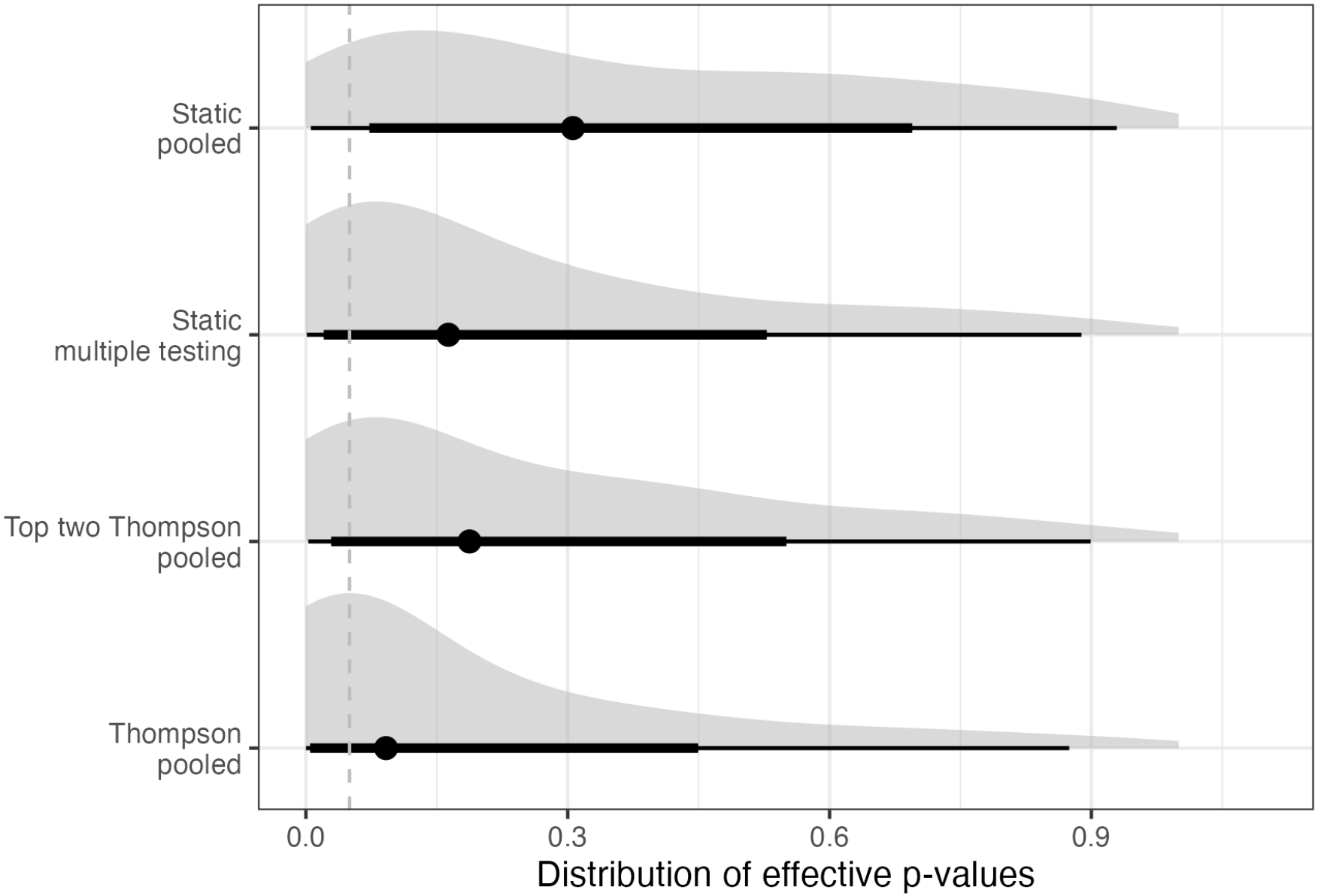

Targeted subgroups

The president's top advisers, by contrast, remain united on the strategy and are pleased by the early responses to the ads among key targeted groups, despite little immediate movement in the polls.

Which ads should the campaign run–for targeted groups?

Bridging non-contextual and contextual settings

- We want to allow that different treatments are best for different types of people.

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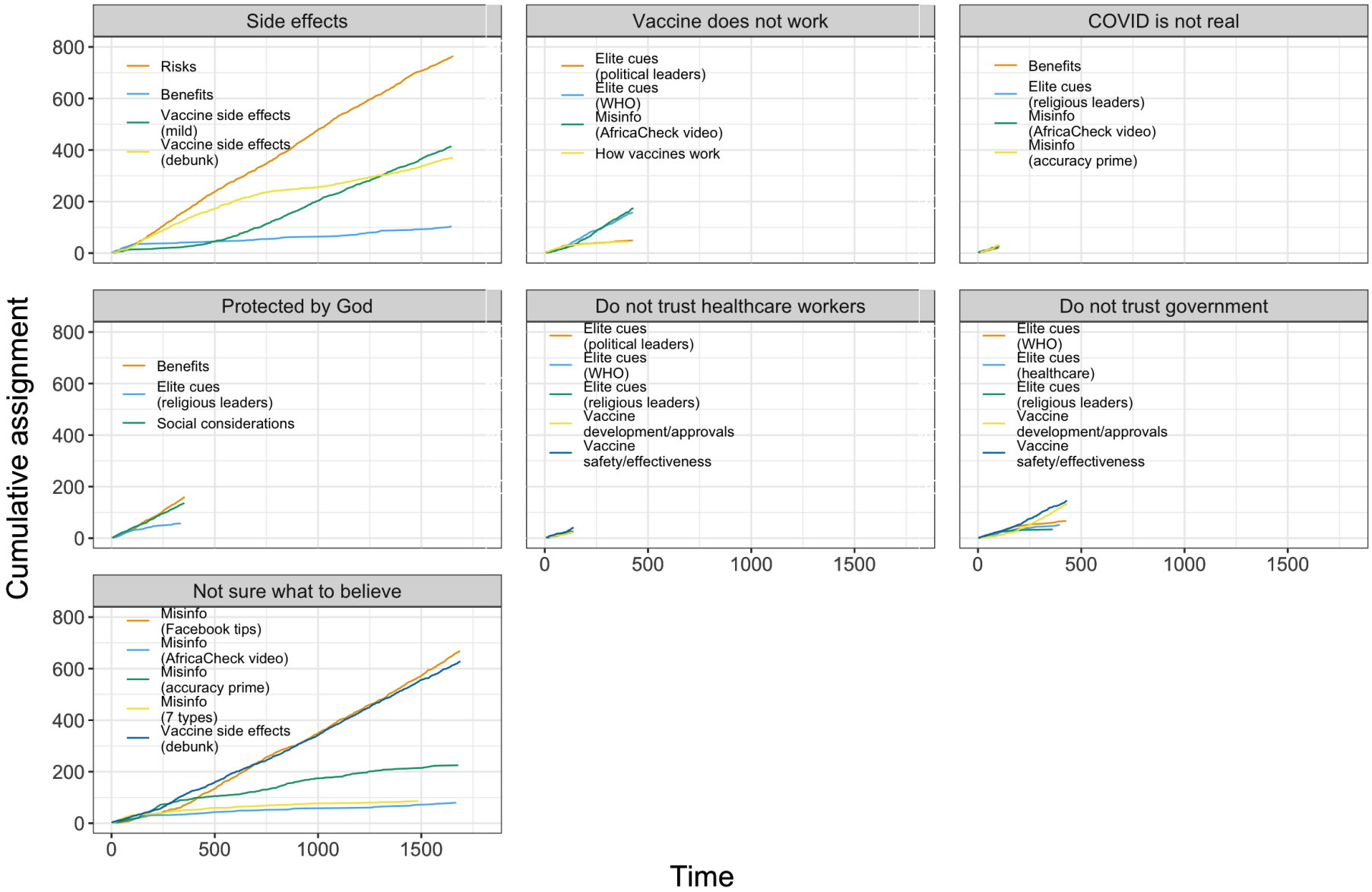
- Assume known, non-overlapping covariate regions.
- Assume no information is shared across regions.

Adaptive assignment



Application I: Vaccine hesitancy

Extending to the fully contextual setting

Fully contextual setting

- We're learning both contexts *and* best context-specific treatments simultaneously.

Planner's goal

$$\pi^* = \arg\max_{\pi \in \Pi} \mathrm{E}\left[\langle \mathbf{Y}_i, \pi(\mathbf{X}_i) \rangle\right]$$

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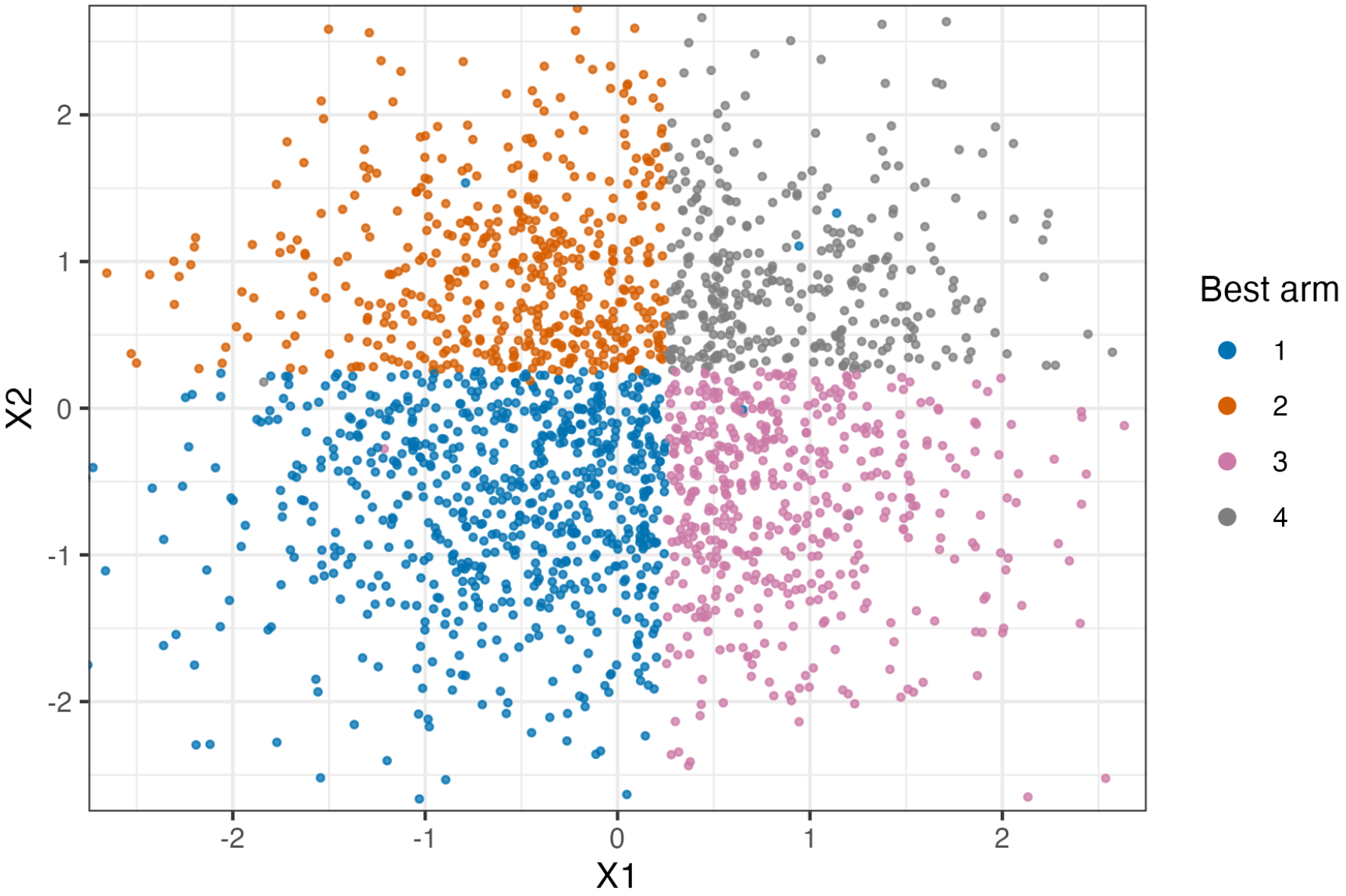
- Contextual **policy learning** (Kitagawa and Tetenov, 2018; Athey and Wager, 2021)
- In practice, π^* is unknowable
- Proximate objective: run an experiment, and *at the end of the experiment*, learn some policy $\hat{\pi}$ to minimize **simple regret**

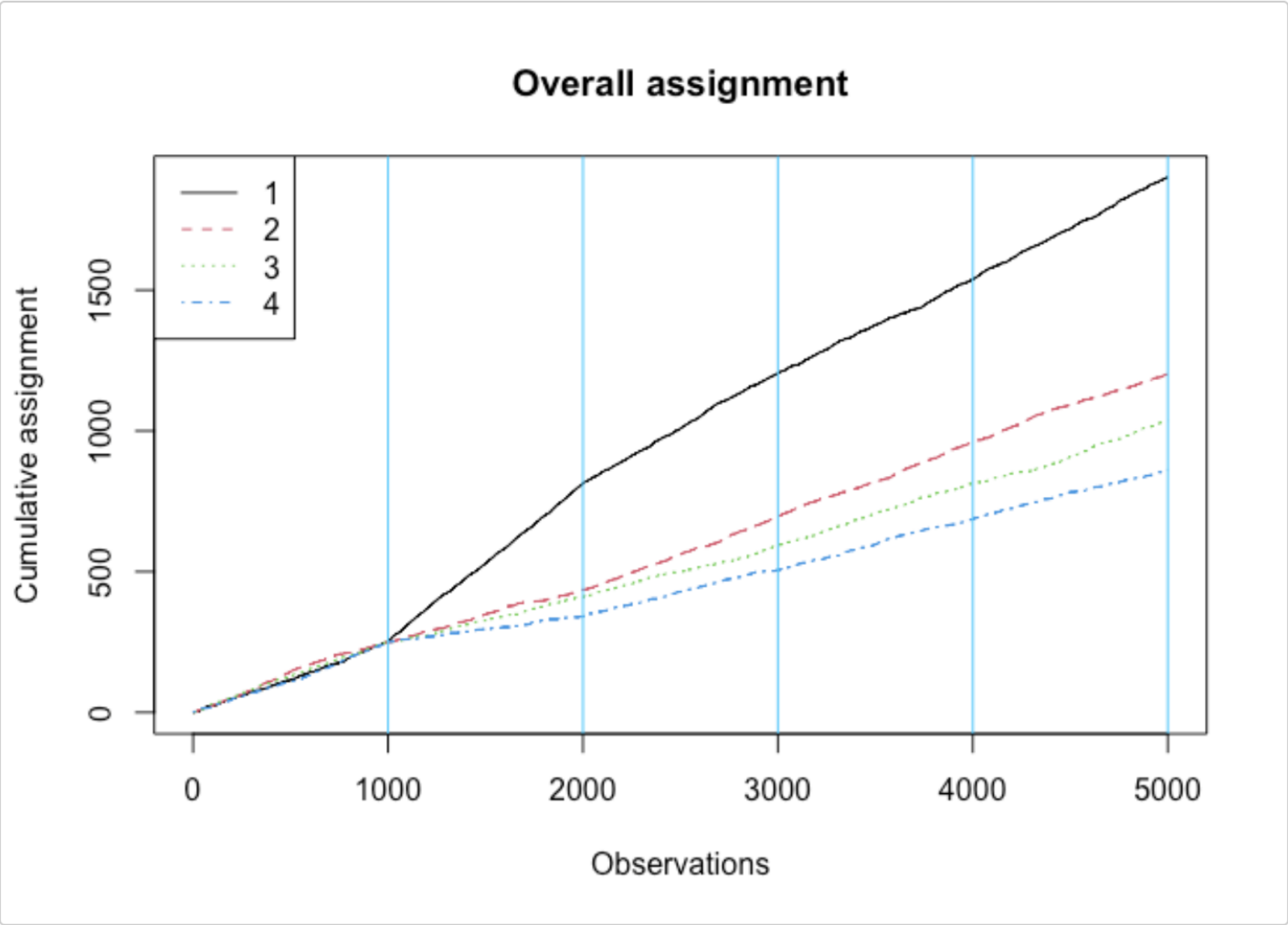
Algorithm Batch-wise balanced linear Thompson sampling

$\Xi \leftarrow$ empty matrix; $\mathbf{X} \leftarrow$ empty matrix; $\mathbf{y} \leftarrow$ empty vector. ▷ Initialize weight matrix, treatment-augmented covariate matrix, and reward vector.
for $i = 1, \ldots, n$ **do**
  if $i \in \mathcal{I}_1$ **then**
    Select $w_i \sim \mathrm{Uniform}(\mathcal{W})$ ▷ In first batch, assign treatment uniformly at random.
  else
    if i is the first observation in $\mathcal{I}_b$ **then** ▷ Update estimates of coefficient vector and variance matrix, using ridge regression with determined penalization.
      $B \leftarrow \mathbf{X}^\top \Xi \mathbf{X} + \lambda^{CV} \mathbf{I}$
      $\hat{\theta} \leftarrow B^{-1} \mathbf{X}^\top \Xi y$
      $\hat{\mathrm{V}}[\hat{\theta}] \leftarrow B^{-1} \left(\left(\mathbf{y} - \mathbf{X}^\top \hat{\theta} \right)^\top \Xi \left(\mathbf{y} - \mathbf{X}^\top \hat{\theta} \right) \right)$
    end if
    for $m = 1, \ldots, M$ **do**
      Sample $\tilde{\theta}^{(m)} \sim \mathcal{N}\left(\hat{\theta}, \hat{\mathrm{V}}[\hat{\theta}]\right)$
    end for
    $q_w \leftarrow \frac{1}{M} \sum_{m=1}^{M} 1 \left\{ w = \arg\max_w \{ x_{[1]i}^\top \tilde{\theta}^{(m)}, \ldots, x_{[|\mathcal{W}|]i}^\top \tilde{\theta}^{(m)} \} \right\}$ ▷ Compute TS probabilities based on observed context.
    $\tilde{q}_w = \max\{q_w, p\} \;\; \forall w \in \mathcal{W}$ ▷ Impose probability floors,
    $u_{\text{total}} = \sum_w \tilde{q}_w - 1$ ▷ and rescale.
    $u_w = \tilde{q}_w - p \;\; \forall w \in \mathcal{W}$
    $c = u_{\text{total}} / \left(\sum_w u_w \right)$
    $e_w = \tilde{q}_w - c * u_w \;\; \forall w \in \mathcal{W}$
  end if
  Select $w_i \sim \mathrm{Multinom}(\mathbf{e})$ ▷ Assign treatment.
  $\xi_i \leftarrow \frac{1}{e_{w_i}}$ ▷ Record inverse probability weights based on realized assignment.
  $\Xi \leftarrow \mathrm{diag}\left(\Xi, \xi_i\right)$ ▷ Augment weight matrix.
  $\mathbf{X} \leftarrow [\mathbf{X} : x_{[w_i]i}^\top]$ ▷ Augment covariate matrix.
  $\mathbf{y} \leftarrow [\mathbf{y} : y_i]$ ▷ Augment reward vector.
end for

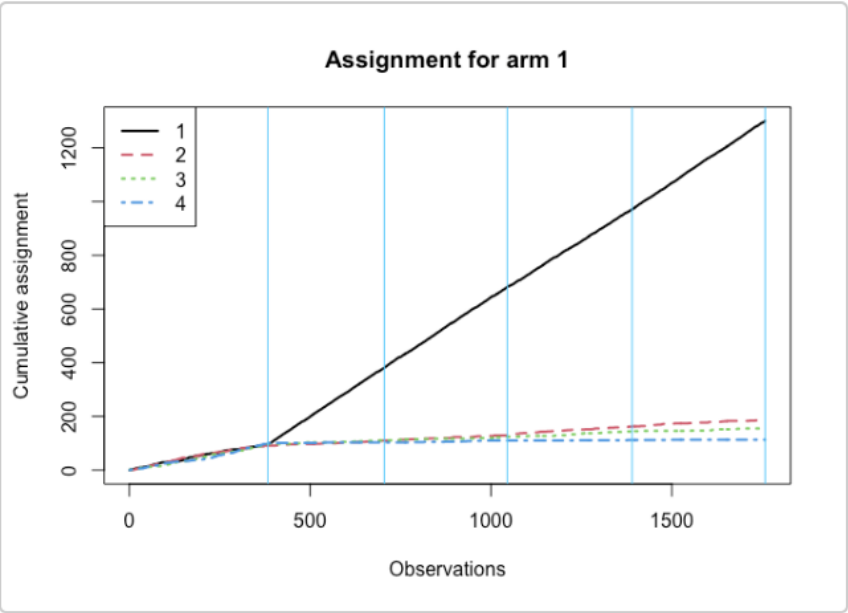
DGP

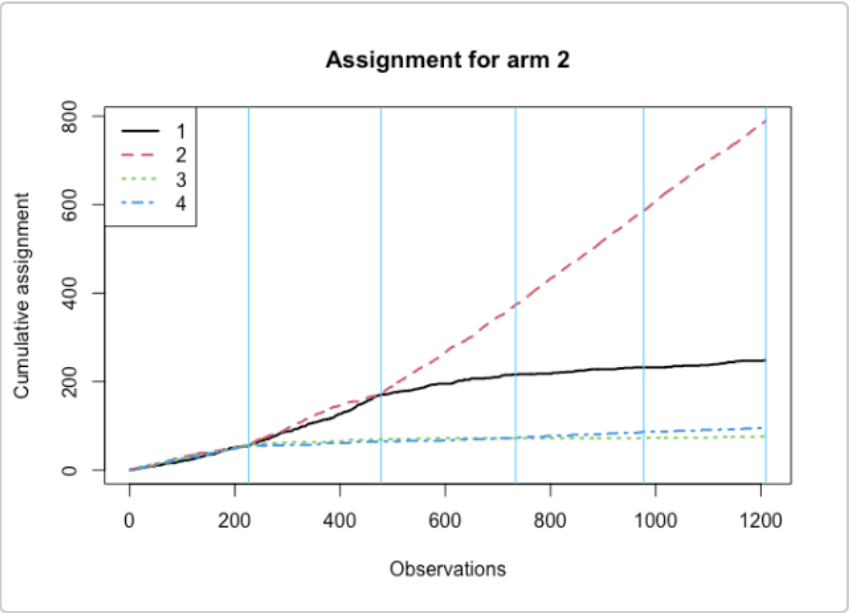


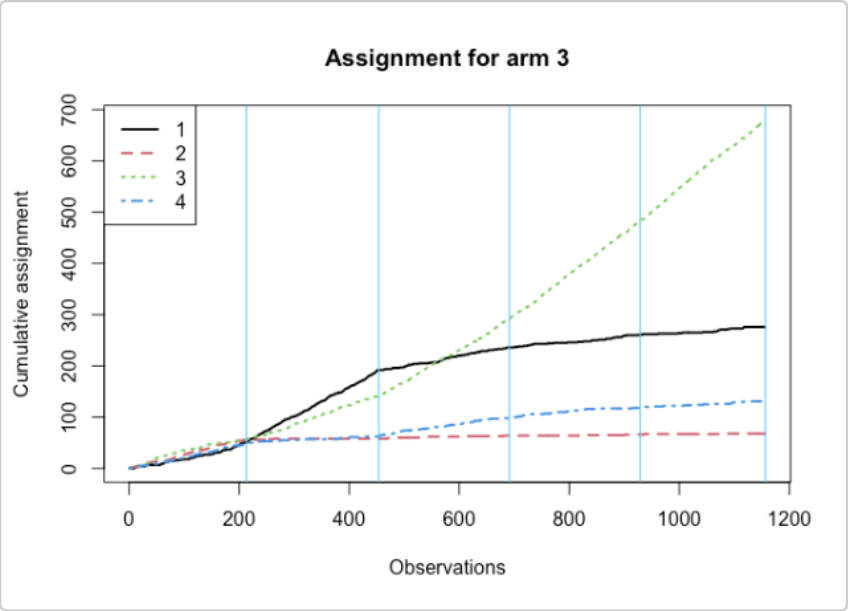
Design

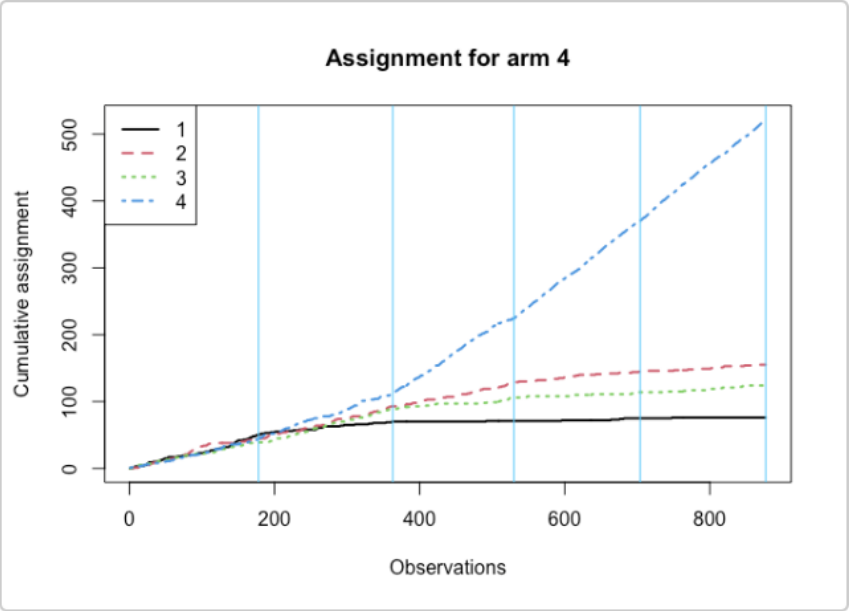

Overall assignment
1
2
3
4
Cumulative assignment
0
500
1000
1500
0
1000
2000
3000
4000
5000
Observations

Design


Assignment for arm 1
Cumulative assignment
Observations
1
2
3
4
0
200
400
600
800
1200
0
500
1000
1500


Assignment for arm 2
Cumulative assignment
Observations
1
2
3
4
0
200
400
600
800
0
200
400
600
800
1000
1200


Assignment for arm 3
Cumulative assignment
Observations
1
2
3
4
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200


Assignment for arm 4
Cumulative assignment
Observations
1
2
3
4
0
100
200
300
400
500
0
200
400
600
800

- `banditsCI`
  https://doi.org/10.5281/zenodo.10516416

- Applications
  - Application I: Vaccine hesitancy
  - Application II: Online misinformation

- Methods
  - Algorithms
  - De-biasing and estimation
  - Contextual recommendation procedures

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Algorithm Batch-wise Bernoulli Thompson sampling

K arms have unknown success rates, $\theta_1, \ldots, \theta_k$, following their respective Bernoulli distributions. Priors take the form of $\mathrm{Be}(\alpha_w, \beta_w)$. Batches are size n.

Initialize priors such that $(\alpha_{w,1} = 1, \beta_{w,1} = 1)$ for $w = 1, \ldots, K$.
for periods $t = 1, \ldots, T$: **do**
    Calculate $e_{w,t} = P\left[\Theta_w = \max\{\Theta_1, \ldots, \Theta_K\} | (\alpha_{1,t}, \beta_{1,t}), \ldots, (\alpha_{K,t}, \beta_{K,t})\right]$ for $w = 1, \ldots, K$. ▷ Calculate posterior probability that each arm is best.
    Sample n observations, assigning treatment with probabilities $(e_{1,t}, \ldots, e_{K,t})$.
    for $w = 1, \ldots, K$ **do** ▷ Update posteriors.
        $\alpha_{w,t+1} \leftarrow \alpha_{w,t} + \#$ successes observed for arm w in period t
        $\beta_{w,t+1} \leftarrow \beta_{w,t} + \#$ failures observed for arm w in period t
    end for
end for

TS explainer

Algorithm Bernoulli top-two Thompson sampling

```
Set α_w ← 1, β_w ← 1 ∀w ∈ 𝒲; some B parameter in (0, 1).                ▷ Initialize priors
for i = 1, . . . , T: do
    for w ∈ 𝒲 do
        Sample θ̂_w ∼ Beta(α_w, β_w)                                      ▷ Sample from posteriors.
    end for
    I ← arg max_w θ_2
    Sample B ∼ Bernoulli(B)
    if B = 1 then
        Assign treatment w_i ← I
    else
        repeat
            for w ∈ 𝒲 do
                Sample θ̂_w ∼ Beta(α_w, β_w)                              ▷ Resample from posteriors.
                J ← arg max_w θ̂_w
            end for
        until J ≠ I
        Assign treatment w_i ← J
    end if
    (α_{w_i}, β_{w_i}) ← (α_{w_i} + 𝟙{y_i = 1} + 1, β_{w_i} + 𝟙{y_i = 0}).   ▷ Update posterior distribution parameters.
end for
```

TTTS explainer

Algorithm Control-augmented Thompson sampling

Suppose some K arms have some unknown success rates, $\theta_1, \ldots, \theta_k$, following their respective Bernoulli distributions. Priors take the form of $\mathrm{Be}(\alpha_w, \beta_w)$. Let batches be of size n.

Let C index the control arm. Initialize priors such that $(\alpha_{w,1} = 1, \beta_{w,1} = 1)$ for $w \neq C$.
for periods $t = 1, \ldots, T$: **do**
  for $w \neq C$ **do**
    Calculate $e_{w,t} = P\left[\Theta_w = \max\{\Theta_1, \ldots, \Theta_K\} | (\alpha_{1,t}, \beta_{1,t}), \ldots, (\alpha_{K,t}, \beta_{K,t})\right]$ for $w = 1, \ldots, K$. ▷ Calculate posterior probability that each arm is best.
  end for
  $b \leftarrow \arg\max_w p_{w,t}$ ▷ Retrieve the "current best arm"...
  $d \leftarrow n_{b,t} - n_{C,t}$ ▷ ...and calculate the difference between the cumulative sample assigned to that arm and the control arm.
  $q \leftarrow \min(\max(d/n, 0), Z_t)$. ▷ Calculate the proportion of the next batch needed for the control to match the cumulative sample of the "best" arm, up to a maximum of Z_t of the batch, where $Z_t \in (0, 1)$ may be fixed, or may be data adaptive.
  $\tilde{p}_{C,t} \leftarrow q + R_t * (1 - q)$. ▷ Calculate the probability of assignment to the control condition, a combination of the allocation to match the cumulative sample of the control to the current best arm, and R_t of the remaining probability, for $R_t \in (0, 1)$. This value may be fixed, or may also be data adaptive
  for $w \neq C$ **do**
    $\tilde{p}_{w,t} \leftarrow p_{w,t} * (1 - R_t) * (1 - q)$ ▷ Calculate the probability of assignment to the treatment arms, according to their posterior probabilities, scaled to the remaining sampling probability.
  end for
  Sample n observations, assigning treatment with probabilities $(\tilde{p}_{1,t}, \ldots, \tilde{p}_{C,t}, \ldots, \tilde{p}_{W,t})$.
  for $w \neq C$ **do** ▷ Update posteriors.
    $\alpha_{w,t+1} \leftarrow \alpha_{w,t} + \#$ successes observed for arm w in period t
    $\beta_{w,t+1} \leftarrow \beta_{w,t} + \#$ failures observed for arm w in period t
  end for
end for

design overview

Algorithm Batch-wise balanced linear Thompson sampling

$\Xi \leftarrow$ empty matrix; $\mathbf{X} \leftarrow$ empty matrix; $\mathbf{y} \leftarrow$ empty vector. ▷ Initialize weight matrix, treatment-augmented covariate matrix, and reward vector.
for $i = 1, \ldots, N$ **do**
  if $i \in \mathcal{I}_1$ **then**
    $e_w \leftarrow \frac{1}{|\mathcal{W}|} \ \forall w \in \mathcal{W}$ ▷ In first batch, assign treatment uniformly at random.
  else if $i \in \mathcal{I}_b$ for $b = 2, \ldots, B$ **then**
    if i is the first observation in $\mathcal{I}_b$ **then** ▷ Update estimates of coefficient vector and variance matrix, using ridge regression with determined penalization.
      $B \leftarrow \mathbf{X}^\top \Xi \mathbf{X} + \lambda^{CV} \mathbf{I}$
      $\hat{\theta} \leftarrow B^{-1} \mathbf{X}^\top \Xi y$
      $\hat{\mathbb{V}}[\hat{\theta}] \leftarrow B^{-1} \left(\left(\mathbf{y} - \mathbf{X}^\top \hat{\theta} \right)^\top \Xi \left(\mathbf{y} - \mathbf{X}^\top \hat{\theta} \right) \right)$
    end if
    for $m = 1, \ldots, M$ **do**
      Sample $\tilde{\theta}^{(m)} \sim \mathcal{N}\left(\hat{\theta}, \hat{\mathbb{V}}[\hat{\theta}]\right)$
    end for
    $q_w \leftarrow \frac{1}{M} \sum_{m=1}^{M} 1 \left\{ w = \arg\max_w \{ x_{[1]_i}^\top \tilde{\theta}^{(m)}, \ldots, x_{[|\mathcal{W}|]_i}^\top \tilde{\theta}^{(m)} \} \right\}$ ▷ Compute TS probabilities based on observed context.
    $\tilde{q}_w = \max\{q_w, p\} \ \forall w \in \mathcal{W}$ ▷ Impose probability floors,
    $u_{\text{total}} = \sum_w \tilde{q}_w - 1$ ▷ and rescale.
    $u_w = \tilde{q}_w - p \ \forall w \in \mathcal{W}$
    $c = u_{\text{total}} / \left(\sum_w u_w\right)$
    $e_w = \tilde{q}_w - c * u_w \ \forall w \in \mathcal{W}$
  end if
  Assign $w_i \sim \text{Multinom}(e_1, \ldots, e_{|\mathcal{W}|})$ ▷ Assign treatment.
  $\xi_i \leftarrow \frac{1}{e_{w_i}}$ ▷ Record inverse probability weights based on realized assignment.
  $\Xi \leftarrow \text{diag}\ (\Xi, \xi_i)$ ▷ Augment weight matrix.
  $\mathbf{X} \leftarrow [\mathbf{X} : x_{[w_i]_i}^\top]$ ▷ Augment covariate matrix.
  $\mathbf{y} \leftarrow [\mathbf{y} : y_i]$ ▷ Augment reward vector.
end for

Finite-n unbiasedness of the Horvitz-Thompson estimator

Under the potential outcomes framework:

$$\mathrm{E}\left[\hat{\theta}_w^{HT}\right] = \frac{1}{N}\sum_{i=1}^{N} Y_i(w).$$

We require only independence of potential outcomes and treatment conditional on history, $Y_i(w) \perp\!\!\!\perp W_i | S_i$, which is given by the experimental design.

$$\mathrm{E}\left[\hat{\theta}_w^{HT}\right] = \mathrm{E}\left[\frac{1}{N}\sum_{i=1}^{N} Y_i \frac{\mathbb{1}\{W_i = w\}}{e_i(w; S_i)}\right]$$

$$= \frac{1}{N}\sum_{i=1}^{N} \mathrm{E}\left[Y_i \frac{\mathbb{1}\{W_i = w\}}{e_i(w; S_i)}\right].$$

Considering the ith unit, by the Law of Iterated Expectations,

$$\mathrm{E}\left[Y_i \frac{\mathbb{1}\{W_i = w\}}{e_i(w; S_i)}\right] = \mathrm{E}\left[\mathrm{E}\left[Y_i \frac{\mathbb{1}\{W_i = w\}}{e_i(w; S_i)}\middle| S_i\right]\right]$$

Taking the interior term, $\mathrm{E}\left[Y_i \frac{\mathbb{1}\{W_i=w\}}{e_i(w;S_i)}\middle| S_i\right]$, by definition,

$$= \mathrm{E}\left[\frac{Y_i \mathbb{1}\{W_i = w\}}{\Pr[W_i = w|S_i]}\middle| S_i\right]$$

By the potential outcomes model,

$$= \mathrm{E}\left[Y_i(w) \times \frac{\mathbb{1}\{W_i = w\}}{\Pr[W_i = w|S_i]}\middle| S_i\right]$$

And because $Y_i \perp\!\!\!\perp W_i | S_i$,

$$= \mathrm{E}\left[Y_i(w)|S_i\right] \times \mathrm{E}\left[\frac{\mathbb{1}\{W_i = w\}}{\Pr[W_i = w|S_i]}\middle| S_i\right]$$

$$= \mathrm{E}\left[Y_i(w)|S_i\right] \times \frac{\mathrm{E}\left[\mathbb{1}\{W_i = w\}|S_i\right]}{\Pr[W_i = w|S_i]}$$

$$= \mathrm{E}\left[Y_i(w)|S_i\right] \times \frac{\Pr[W_i = w|S_i]}{\Pr[W_i = w|S_i]}$$

$$= \mathrm{E}\left[Y_i(w)|S_i\right].$$

Then returning to the Law of Iterated Expectations from above,

$$\mathrm{E}\left[Y_i \frac{\mathbb{1}\{W_i = w\}}{e_i(w; S_i)}\right] = \mathrm{E}\left[\mathrm{E}\left[Y_i(w)|S_i\right]\right] = \mathrm{E}\left[Y_i(w)\right].$$

HT estimator HT toy example navigation

De-biasing and estimation

navigation

Bias and adaptively collected data

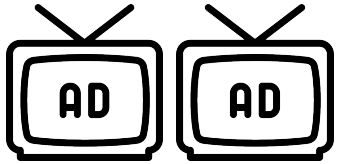
- If we ignore method of data generation and use sample means, adaptively collected data are going to give us estimates that are systematically negatively biased (Nie et al., 2017).

Bias in naive estimation w/ adaptive design

Treatments	A	B
(Unknown) mean response	0.5	0.5

Toy example

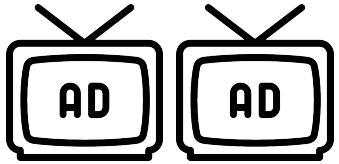
Bias in naive estimation w/ adaptive design

Treatments	
	A **B**
(Unknown) mean response	0.5 0.5

Toy example

- **Period 1:** Assign **one subject each** to treatment A, treatment B.

Bias in naive estimation w/ adaptive design

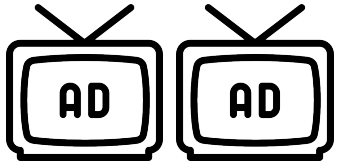
Treatments	
	A **B**
(Unknown) mean response	0.5 0.5

Toy example

- **Period 1:** Assign **one subject each** to treatment A, treatment B.

→ Observe data.

Bias in naive estimation w/ adaptive design

Treatments		
	A	**B**
(Unknown) mean response	0.5	0.5

Toy example

- **Period 1:** Assign **one subject each** to treatment A, treatment B.

→ Observe data.

- **Period 2:** Assign treatment probabilistically to **one subject** using Thompson sampling.

Bias in naive estimation w/ adaptive design

Treatments	**A**	**B**
(Unknown) mean response	0.5	0.5

Toy example

- **Period 1:** Assign **one subject each** to treatment A, treatment B.

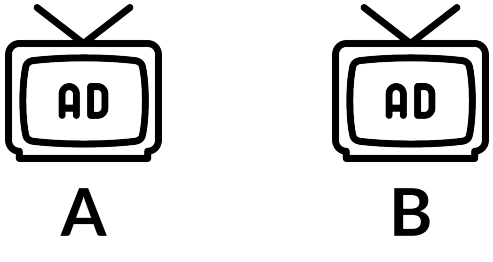
→ Observe data.

- **Period 2:** Assign treatment probabilistically to **one subject** using Thompson sampling.

→ Three subjects total, two periods. Estimate sample means.

Bias in naive estimation w/ adaptive design: one realized experiment

Treatments



Bias in naive estimation w/ adaptive design: one realized experiment

Treatments	**A**	**B**
Period 1		
- sample	A	B

Bias in naive estimation w/ adaptive design: one realized experiment

Treatments	**A**	**B**
Period 1		
- sample	A	B
- outcome	1	0

Bias in naive estimation w/ adaptive design: one realized experiment

Treatments	**A**	**B**
Period 1		
- sample	A	B
- outcome	1	0
- posterior distributions	$\mathrm{Be}(2,1)$	$\mathrm{Be}(1,2)$

Bias in naive estimation w/ adaptive design: one realized experiment

Treatments	**A**	**B**
Period 1		
- sample	A	B
- outcome	1	0
- posterior distributions	$\mathrm{Be}(2,1)$	$\mathrm{Be}(1,2)$
Sampling probability	5/6	1/6

Bias in naive estimation w/ adaptive design: one realized experiment

Treatments	**A**	**B**
Period 1		
- sample	A	B
- outcome	1	0
- posterior distributions	$\mathrm{Be}(2,1)$	$\mathrm{Be}(1,2)$
Sampling probability	5/6	1/6
Period 2		
- sample	A	-

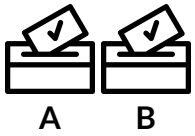
Bias in naive estimation w/ adaptive design: one realized experiment

Treatments	**A**	**B**
Period 1		
- sample	A	B
- outcome	1	0
- posterior distributions	$\mathrm{Be}(2,1)$	$\mathrm{Be}(1,2)$
Sampling probability	5/6	1/6
Period 2		
- sample	A	-
- outcome	0	-

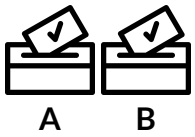
Bias in naive estimation w/ adaptive design: one realized experiment

Treatments	**A**	**B**
Period 1		
- sample	A	B
- outcome	1	0
- posterior distributions	$\mathrm{Be}(2,1)$	$\mathrm{Be}(1,2)$
Sampling probability	5/6	1/6
Period 2		
- sample	A	-
- outcome	0	-
Sample mean	0.5	0

Bias in naive estimation w/ adaptive design: all paths



Bias in naive estimation w/ adaptive design: all paths



Period 1

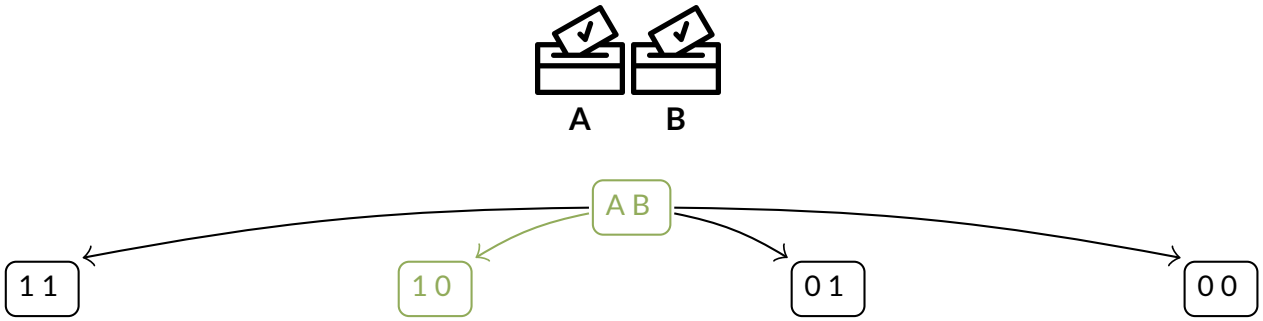
- sample

A B

Bias in naive estimation w/ adaptive design: all paths

Period 1

- sample
- outcome



Bias in naive estimation w/ adaptive design: all paths

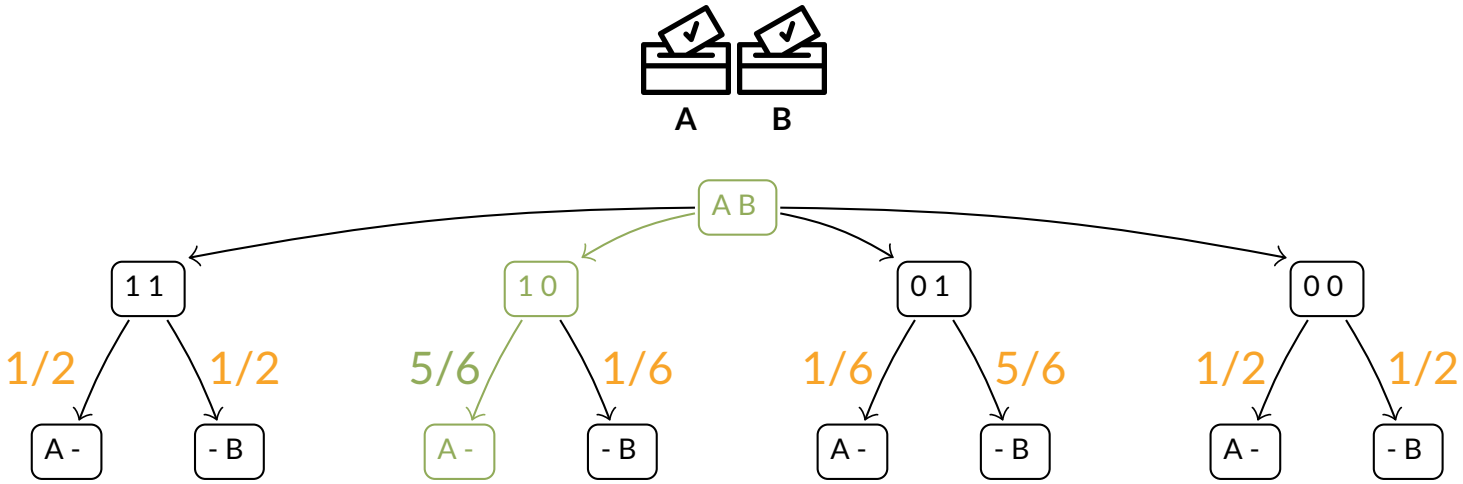
Period 1

- sample

- outcome

Period 2

- sample



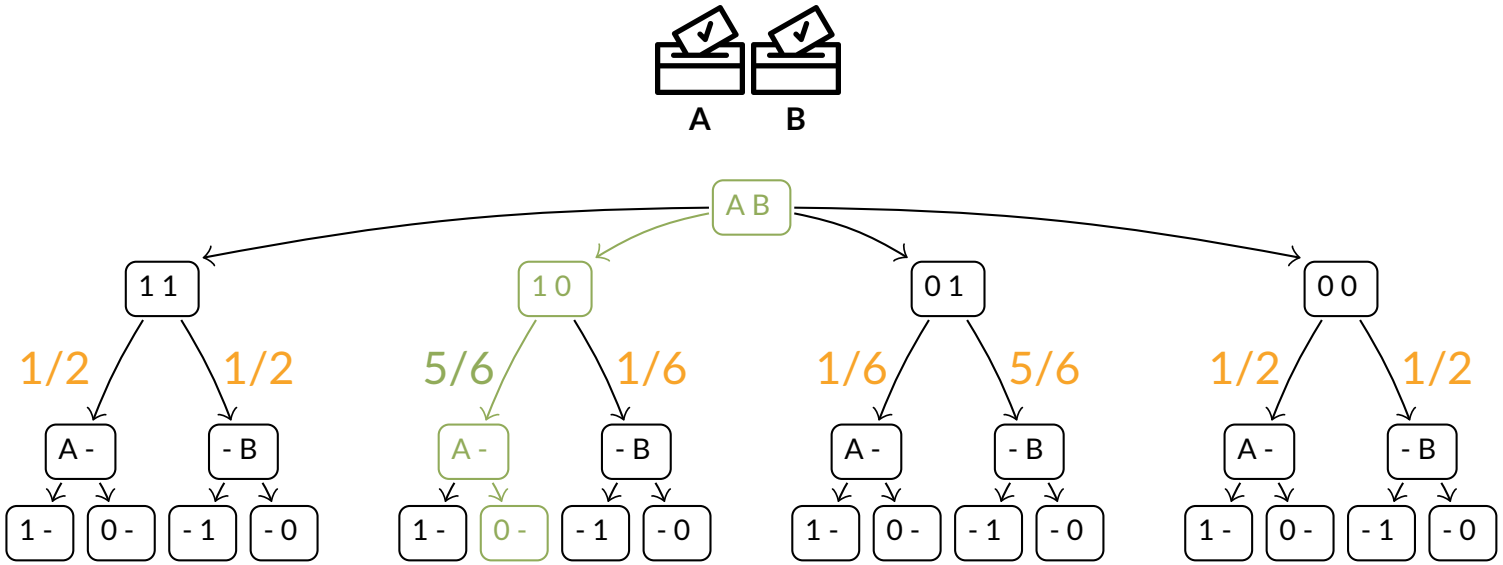
Bias in naive estimation w/ adaptive design: all paths

Period 1

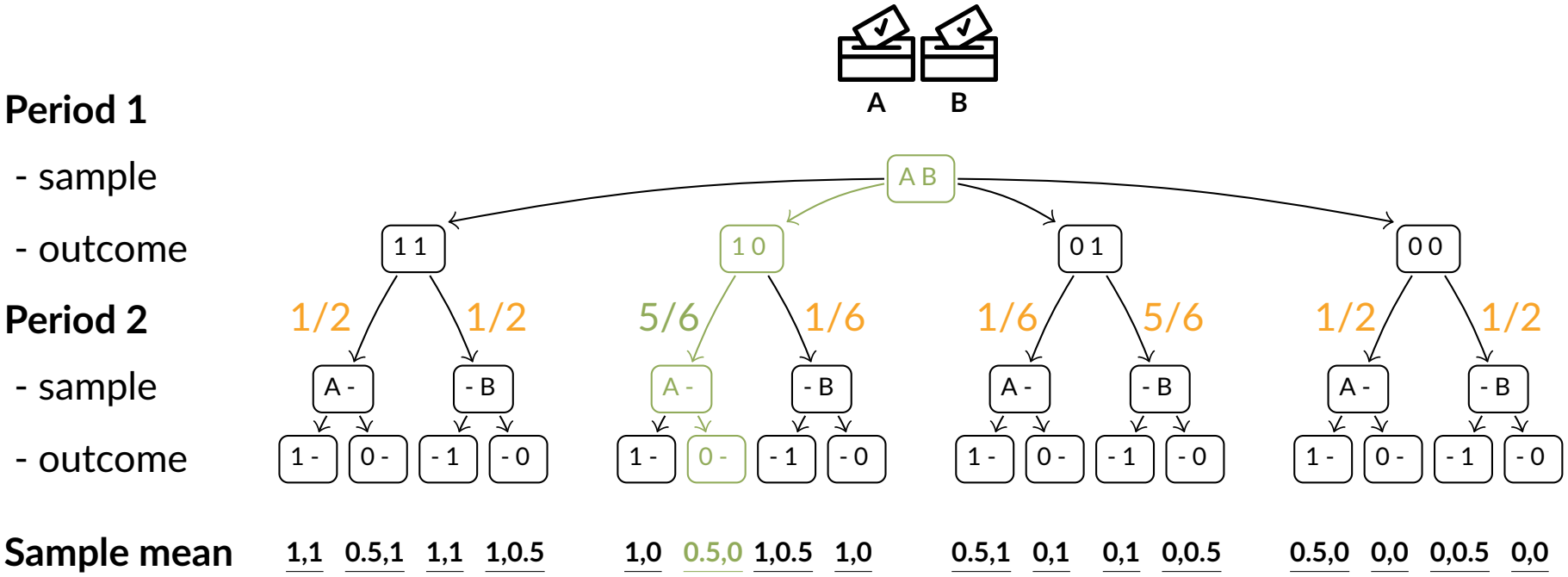
- sample
- outcome

Period 2

- sample
- outcome

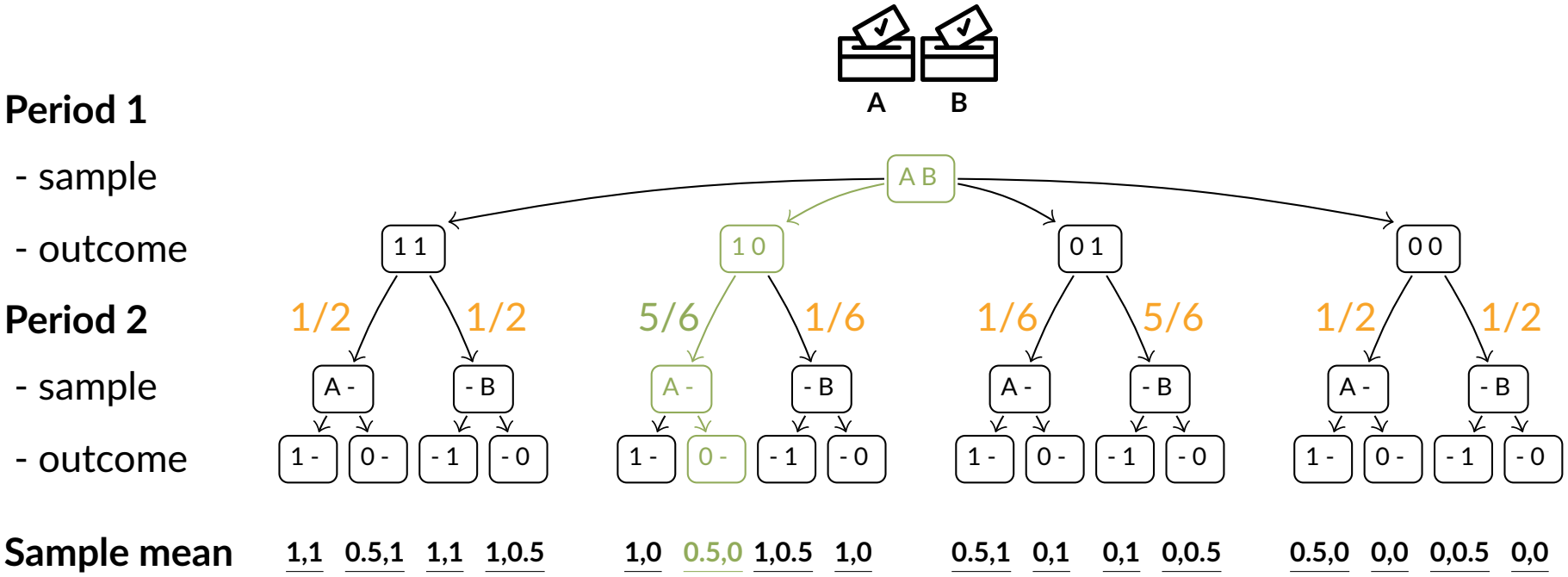


Bias in naive estimation w/ adaptive design: all paths



Bias in naive estimation w/ adaptive design: all paths

After observing three outcomes, bias is negative.



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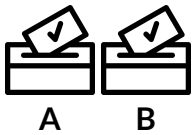


Period 1

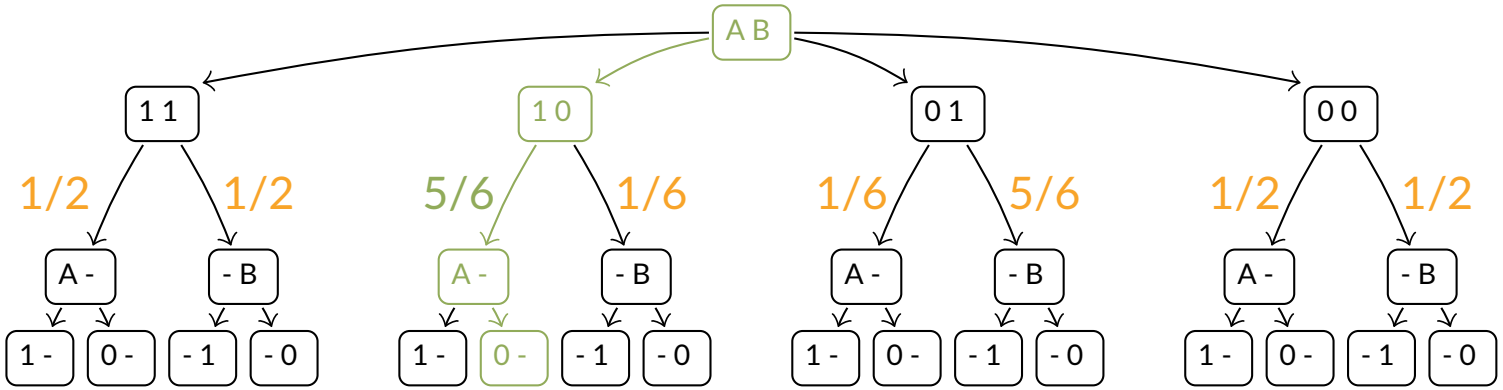
- sample

- outcome

Period 2

- sample

- outcome



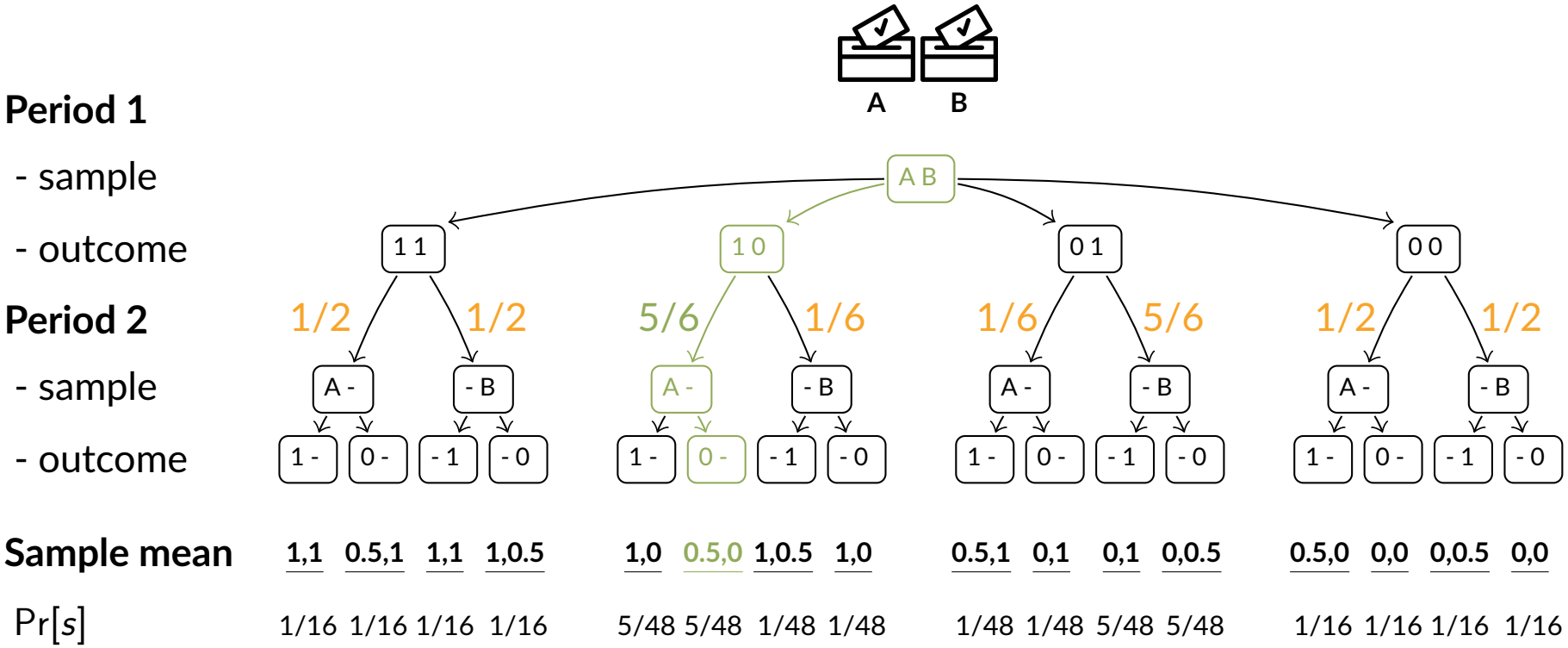
Sample mean	1,1	0.5,1	1,1	1,0.5	1,0	0.5,0	1,0.5	1,0	0.5,1	0,1	0,1	0,0.5	0.5,0	0,0	0,0.5	0,0
Pr[s]	1/16	1/16	1/16	1/16	5/48	5/48	1/48	1/48	1/48	1/48	5/48	5/48	1/16	1/16	1/16	1/16

$$\mathrm{E}\left[\hat{\theta}_w\right] = \sum_s \hat{\theta}_w | s \cdot \Pr[s]$$

$$\approx 0.458$$

Bias in naive estimation w/ adaptive design: all paths

After observing three outcomes, bias is negative.



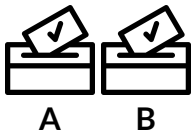
$$\mathrm{E}\left[\hat{\theta}_w\right] = \sum_s \hat{\theta}_w | s \cdot \Pr[s]$$

$$\approx 0.458$$

(S_i is a state, defined by the history of treatment and outcomes.)

Bias in naive estimation w/ adaptive design: all paths

After observing three outcomes, bias is negative.

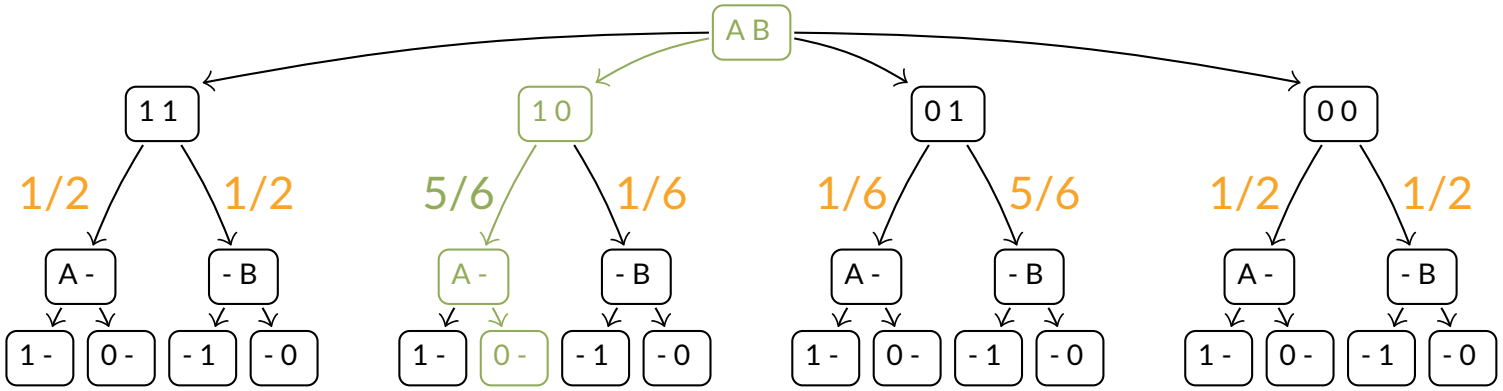


Period 1

- sample
- outcome

Period 2

- sample
- outcome



Sample mean	**1,1**	**0.5,1**	**1,1**	**1,0.5**	**1,0**	**0.5,0**	**1,0.5**	**1,0**	**0.5,1**	**0,1**	**0,1**	**0,0.5**	**0.5,0**	**0,0**	**0,0.5**	**0,0**
Pr[s]	1/16	1/16	1/16	1/16	5/48	5/48	1/48	1/48	1/48	1/48	5/48	5/48	1/16	1/16	1/16	1/16

Why? Lower probability of sampling from an arm if it performs poorly in the first period.

We can de-bias with weighting estimators

Inverse Probability Weighting estimation (Horvitz and Thompson, 1952)

$$\hat{\theta}_w^{HT} = \frac{1}{N} \sum_{i=1}^{N} Y_i \frac{\mathbb{1}\{W_i = w\}}{e_i(w; S_i)}$$

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- (Conditional on algorithm) $e_i(w; S_i) > 0, \forall i, w$.

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- Finite-n unbiased (Bowden and Trippa, 2017) Finite-n unbiasedness

We can de-bias with weighting estimators

Inverse Probability Weighting estimation (Horvitz and Thompson, 1952)

$$\hat{\theta}_w^{HT} = \frac{1}{N} \sum_{i=1}^{N} Y_i \frac{\mathbb{1}\{W_i = w\}}{e_i(w; S_i)}$$

- (Conditional on algorithm) $e_i(w; S_i) > 0, \forall i, w$.
- Finite-n unbiased (Bowden and Trippa, 2017) Finite-n unbiasedness
- Drawback: if there is low overlap between policy you want to evaluate, and the assigned policy, high variance. One of the tradeoffs we face with adaptively collected data.

Estimation with Inverse Probability Weighting

$$\hat{\theta}_w^{HT} = \frac{1}{N} \sum_{i=1}^{N} Y_i \frac{\mathbb{1}\{W_i = w\}}{e_i(w; S_i)}$$

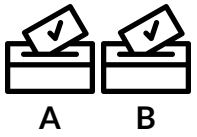
Estimation with Inverse Probability Weighting

$$\hat{\theta}_w^{HT} = \frac{1}{N} \sum_{i=1}^{N} Y_i \frac{\mathbb{1}\{W_i = w\}}{e_i(w; S_i)}$$

In our example…

$$\begin{aligned}\hat{\theta}_A^{HT} &= \frac{1}{3}\left[1 \times \frac{\mathbb{1}\{W_i = A\}}{1/2} + 0 \times \frac{\mathbb{1}\{W_i = A\}}{5/6}\right] \\ &= \frac{2}{3}\end{aligned}$$

De-biasing estimation: all paths

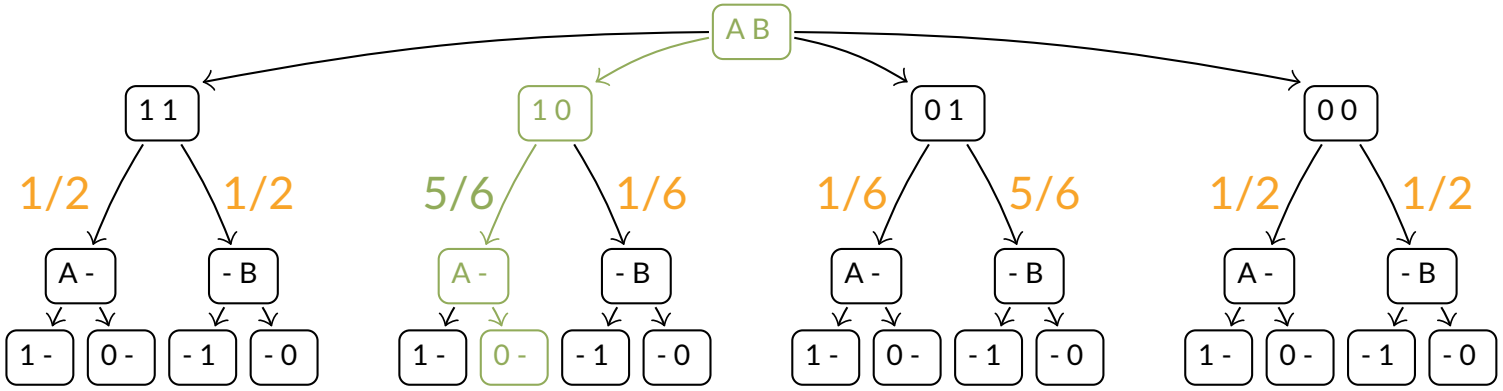


Period 1

- sample
- outcome

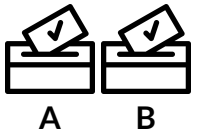
Period 2

- sample
- outcome



Sample mean	**1,1**	**0.5,1**	**1,1**	**1,0.5**	**1,0**	**0.5,0**	**1,0.5**	**1,0**	**0.5,1**	**0,1**	**0,1**	**0,0.5**	**0.5,0**	**0,0**	**0,0.5**	**0,0**
Pr[*s*]	1/16	1/16	1/16	1/16	5/48	5/48	1/48	1/48	1/48	1/48	5/48	5/48	1/16	1/16	1/16	1/16

De-biasing estimation: all paths

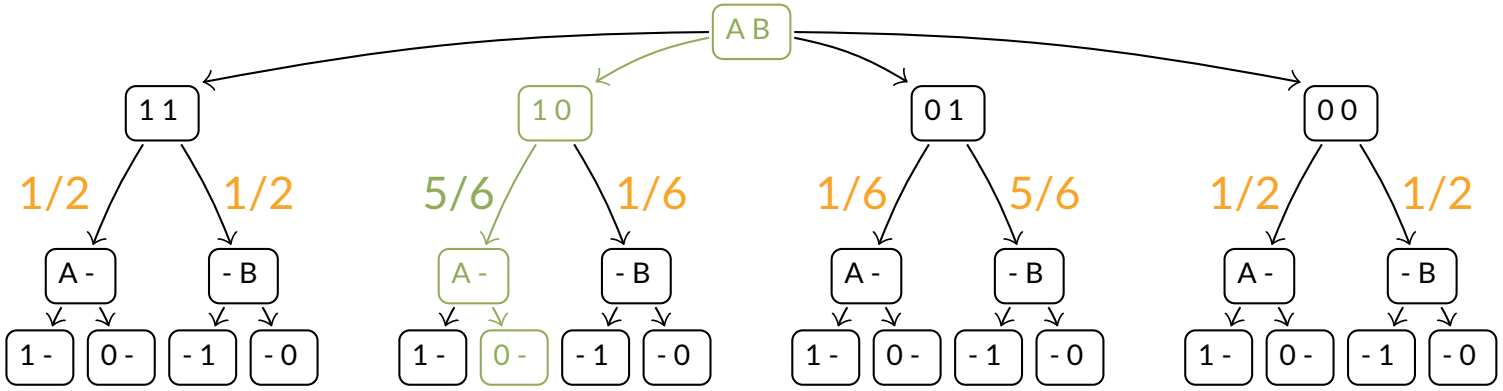


Period 1

- sample
- outcome

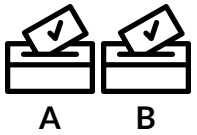
Period 2

- sample
- outcome



HT estimate	$\frac{4}{3},\frac{2}{3}$	$\frac{2}{3},\frac{2}{3}$	$\frac{2}{3},\frac{4}{3}$	$\frac{2}{3},\frac{2}{3}$	$\frac{16}{15},0$	$\frac{2}{3},0$	$\frac{2}{3},2$	$\frac{2}{3},0$	$2,\frac{2}{3}$	$0,\frac{2}{3}$	$0,\frac{16}{15}$	$0,\frac{2}{3}$	$\frac{2}{3},0$	$0,0$	$0,\frac{2}{3}$	$0,0$
Pr[s]	1/16	1/16	1/16	1/16	5/48	5/48	1/48	1/48	1/48	1/48	5/48	5/48	1/16	1/16	1/16	1/16

De-biasing estimation: all paths

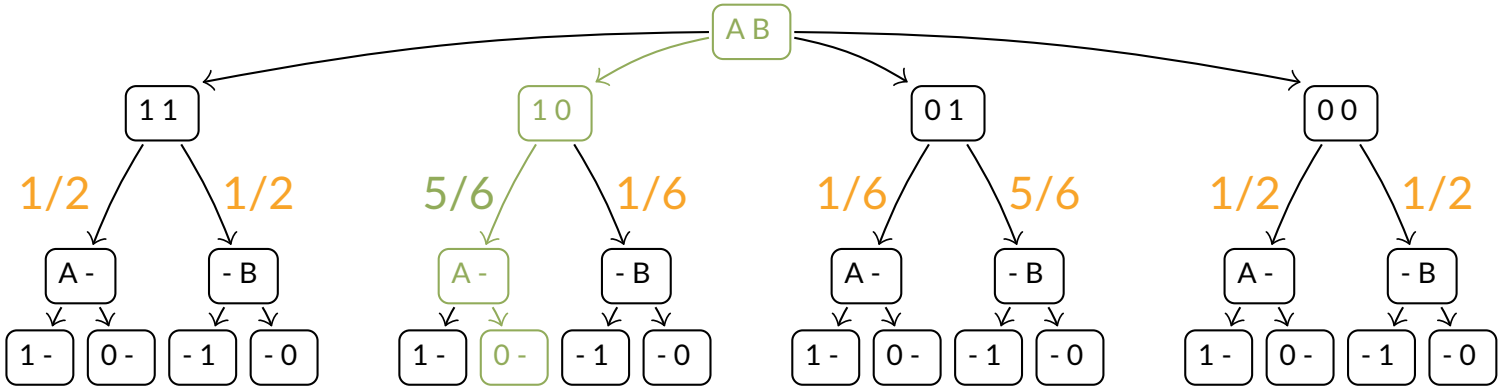


Period 1

- sample
- outcome

Period 2

- sample
- outcome



HT estimate	$\frac{4}{3},\frac{2}{3}$	$\frac{2}{3},\frac{2}{3}$	$\frac{2}{3},\frac{4}{3}$	$\frac{2}{3},\frac{2}{3}$	$\frac{16}{15},0$	$\frac{2}{3},0$	$\frac{2}{3},2$	$\frac{2}{3},0$	$2,\frac{2}{3}$	$0,\frac{2}{3}$	$0,\frac{16}{15}$	$0,\frac{2}{3}$	$\frac{2}{3},0$	$0,0$	$0,\frac{2}{3}$	$0,0$
Pr[s]	1/16	1/16	1/16	1/16	5/48	5/48	1/48	1/48	1/48	1/48	5/48	5/48	1/16	1/16	1/16	1/16

$$\mathrm{E}\left[\hat{\theta}_w^{HT}\right] = \sum_s \hat{\theta}_w^{HT}|s \cdot \Pr[s]$$

$$= 0.5$$

HT unbiased

Covariate adjustment: doubly robust estimation

We can also consider inverse probability weighting augmented by covariate adjustment.

$$\hat{\theta}_w^{DR} = \frac{1}{N}\sum_{i=1}^{N} \underbrace{\hat{\mathrm{E}}\left[Y_i | W_i = w, X_i\right]}_{\text{estimated outcome model}} + \frac{1}{N}\sum_{i=1}^{N} \underbrace{\frac{\mathbb{1}\{W_i = w\}\left(Y_i - \hat{\mathrm{E}}\left[Y_i | W_i = w, X_i\right]\right)}{e_i(w; s)}}_{\text{residual bias correction}}$$

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- We can use regression–or machine-learning methods–for estimation of the outcome model,

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